Sanchit Jain

Legend

I am a computer science faculty, having a teaching experience of more than 8 years. I have a very sound experience in teaching students.

46M Watch mins | 3M Watch mins (last 30 days) | 21K Followers | 2K Dedications

Educator highlights

- Works at **Knowledge Gate**
- Studied at **Delhi Technological University**
- Have qualified gate Have experience of more than 8 years Have a YouTube channel with 5 lakh follower
- Lives in **Ghaziabad, Uttar Pradesh, India**
- Unacademy Educator since **19th July, 2019**
- **844,388 live minutes** taught in last 30 days
- Knows **Punjabi, Hinglish, Hindi and English**

Months Wise Calendar for GATE-2022 (Unacademy Plus) (Sanchit Jain)

Month	Morning Class (7:00am – 9:00am)	Evening Class (7:00pm – 9:00pm)
Feb-2021	DS & Programming (15-Feb to 23-Mar) (56 Hours)	
Mar-2021		DM (22-Feb to 27-Mar) (50 Hours)
Apr-2021	Algo (24-Mar to 26-April) (38 Hours)	DBMS (29-Mar to 30-April) (50 Hours)
May-2021	CN(29-April to 15-June) (54 Hours)	DE (7-May to 1-June) (40 Hours)
Jun-2021	TOC (26-May to 05-July) (58 Hours)	
Jul-2021	Compiler (7-July to 5-Aug) (34 Hours)	COA (23-June to 20-July) (38 Hours)
Aug-2021		OS (21-July to 24-Aug) (58 Hours)
Sep-2021	DM (11-Aug to 20-Sept) (50 Hours)	DS & Programming (25-Aug to 5-Oct) (56 Hours)
Oct-2021	DBMS (22-Sept to 29-Oct) (50 Hours)	Algo (6-Oct to 1-Nov) (38 Hours)
Nov-2021	DE (3-Nov to 23-Nov) (40 Hours)	**CN(10-Nov to 5 Jan) (54 Hours)**
Dec-2021	**COA (24-Nov to 13-Dec) (38 Hours)**	
Jan-2022	OS (15-Dec to 11-Jan) (58 Hours)	Compiler (26-Jan to 15-Feb) (34 Hours)
Feb-2022	TOC (12-Jan to 12-Feb) (58 Hours)	

Date	Day	Subject-1	Topic	Time
24-11-2021	Wednesday	COA	memory hierarchy	7:00 am to 9:00 am
25-11-2021	Thursday	COA	Cache Mapping Part-1	7:00 am to 9:00 am
26-11-2021	Friday	COA	Cache Mapping Part-2	7:00 am to 9:00 am
27-11-2021	Saturday	COA	Doubt Clearing Session	7:00 am to 9:00 am
28-11-2021	Sunday	COA	Cache Management Part-1	7:00 am to 9:00 am
29-11-2021	Monday	COA	Cache Management Part-2	7:00 am to 9:00 am
30-11-2021	Tuesday			
01-12-2021	Wednesday	COA	Input-Output Organization Part-1	7:00 am to 9:00 am
02-12-2021	Thursday	COA	Doubt Clearing Session	7:00 am to 9:00 am
03-12-2021	Friday	COA	Input-Output Organization Part-2	7:00 am to 9:00 am
04-12-2021	Saturday	COA	Input-Output Organization Part-3	7:00 am to 9:00 am
05-12-2021	Sunday	COA	Input-Output Organization Part-4	7:00 am to 9:00 am
06-12-2021	Monday	COA	Doubt Clearing Session	7:00 am to 9:00 am
07-12-2021	Tuesday	COA	Pipelining Part-1	7:00 am to 9:00 am
08-12-2021	Wednesday	COA	Pipelining Part-2	7:00 am to 9:00 am
09-12-2021	Thursday	COA	Instruction Formate and Addressing mode Part-1	7:00 am to 9:00 am
10-12-2021	Friday	COA	Doubt Clearing Session	7:00 am to 9:00 am
11-12-2021	Saturday	COA	Instruction Formate and Addressing mode Part-2	7:00 am to 9:00 am
12-12-2021	Sunday	COA	Control Unit Part-1	7:00 am to 9:00 am
13-12-2021	Monday	COA	Control Unit Part-2	7:00 am to 9:00 am

Date	Day	Subject-2	Topic	Time	Duration
10-11-2021	Wednesday	CN	Basics of Computer Networks Part-1	6:00 pm to 8:00 pm	2
11-11-2021	Thursday	CN	Basics of Computer Networks Part-2	6:00 pm to 8:00 pm	2
15-11-2021	Monday	CN	Basics of Computer Networks Part-3	6:00 pm to 8:00 pm	2
16-11-2021	Tuesday	CN	Doubt Clearing Session	6:00 pm to 8:00 pm	2
17-11-2021	Wednesday	CN	Access Control Part-1	6:00 pm to 8:00 pm	2
18-11-2021	Thursday	CN	Access Control Part-2	6:00 pm to 8:00 pm	2
22-11-2021	Monday	CN	Flow Control Part-1	6:00 pm to 8:00 pm	2
23-11-2021	Tuesday	CN	Doubt Clearing Session	6:00 pm to 8:00 pm	2
24-11-2021	Wednesday	CN	Error Control Part-1	6:00 pm to 8:00 pm	2
25-11-2021	Thursday	CN	Error Control Part-2	6:00 pm to 8:00 pm	2
29-11-2021	Monday	CN	Framing and Ethernet	6:00 pm to 8:00 pm	2
30-11-2021	Tuesday	CN	Doubt Clearing Session	6:00 pm to 8:00 pm	2
01-12-2021	Wednesday	CN	Network Layer protocol Part-1	6:00 pm to 8:00 pm	2
02-12-2021	Thursday	CN	Network Layer protocol Part-2	6:00 pm to 8:00 pm	2
06-12-2021	Monday	CN	Network Layer protocol Part-3	6:00 pm to 8:00 pm	2
07-12-2021	Tuesday	CN	Doubt Clearing Session	6:00 pm to 8:00 pm	2
13-12-2021	Monday	CN	IP addressing and routing and Routing Part-1	6:00 pm to 8:00 pm	2
14-12-2021	Tuesday	CN	IP addressing and routing and Routing Part-2	6:00 pm to 8:00 pm	2
20-12-2021	Monday	CN	Transport Layer Part-1	6:00 pm to 8:00 pm	2
21-12-2021	Tuesday	CN	Doubt Clearing Session	6:00 pm to 8:00 pm	2
27-12-2021	Monday	CN	Transport Layer Part-2	6:00 pm to 8:00 pm	2
28-12-2021	Tuesday	CN	Transport Layer Part-3	6:00 pm to 8:00 pm	2
03-01-2022	Monday	CN	Application Layer	6:00 pm to 8:00 pm	2
04-01-2022	Tuesday	CN	Doubt Clearing Session	6:00 pm to 8:00 pm	2
05-01-2022	Wednesday	CN	crypto	6:00 pm to 8:00 pm	2

GATE & ESE | Plus | All courses | All languages | All topics

HINGLISH ELECTRICAL ENGINEERING

Complete Course on Computer Architecture for GATE

Lesson 7 • Dec 1, 2021 7:00 AM

Sanchit Jain

HINGLISH CS & IT

Complete Course on Computer Networks for GATE

Lesson 11 • Today, 6:00 PM

Sanchit Jain

HINDI CS & IT

Complete Course on Digital Electronics for GATE

Ended on Nov 22, 2021 • 18 lessons

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HINGLISH PRACTICE & STRATEGY

FAQ Handbook for GATE

Ended on Nov 26, 2021 • 7 lessons

Ravindrababu Ravula

HINDI CS & IT

Complete Course on Algorithms for GATE

Ended on Nov 9, 2021 • 19 lessons

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Complete Course on DBMS for GATE

Ended on Oct 28, 2021 • 25 lessons

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Complete Course on Programming & Data Structures for GATE

Ended on Oct 12, 2021 • 28 lessons

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Complete Course on Discrete Mathematics for GATE

Ended on Oct 10, 2021 • 27 lessons

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Ended on Aug 23, 2021 • 30 lessons

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Complete Course on Compiler Design for GATE

Ended on Aug 13, 2021 • 18 lessons

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HINDI CS & IT

Complete Course on Computer Architecture for GATE 2022

Ended on Jul 20, 2021 • 19 lessons

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HINGLISH CS & IT

Foundation Course on Theory of Computation for GATE 2022

Ended on Jul 5, 2021 • 29 lessons

Sanchit Jain

Cache Replacement Policies

- In direct mapped cache, the position of each block is predetermined hence no replacement policy exists.

- In fully associative and set associative caches there exists policies.

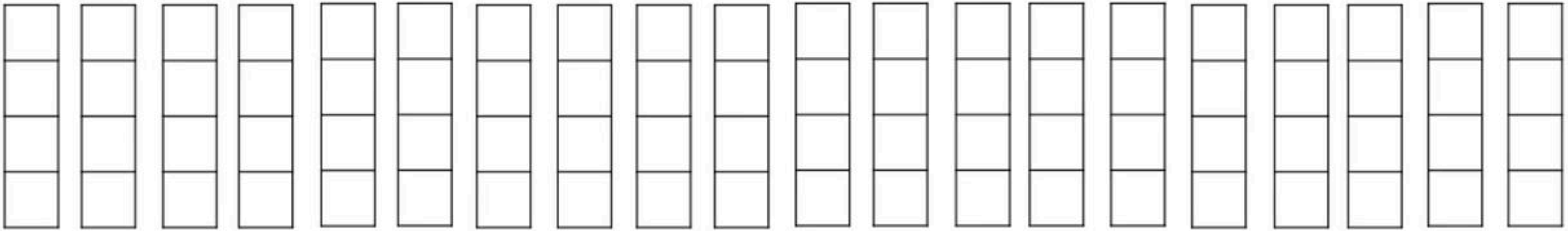
- When a new block is brought into the cache and all the positions that it may occupy are full, then the controller needs to decide which of the old blocks it can overwrite.

FIFO Policy

- The block which have entered first in the memory will be replaced first.
- This can lead to a problem known as "**Belady's Anomaly**", it states that if we increase the number of lines in cache memory the cache miss will increase.

Example: Let the blocks be in the sequence: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 and the cache memory has 4 lines.

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1



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Example: Let the blocks be in the sequence: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 and the cache memory has 4 lines.

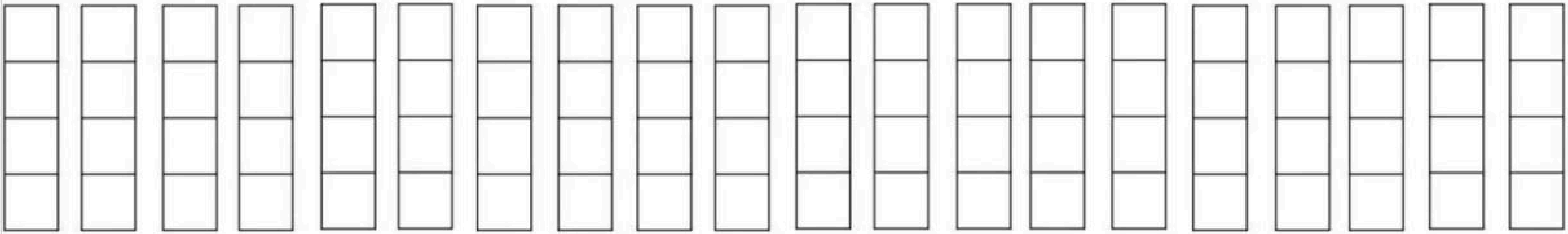
			2	2	2	2	1	1	1
		1	1	1	1	0	0	0	0
	0	0	0	0	4	4	4	4	7
7	7	7	7	3	3	3	3	2	2

LRU (Least Recently Used)

- The page which was not used for the longest period of time in the past will get replaced first.

Example: Let the blocks be in the sequence: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 and the cache memory has 4 lines.

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1



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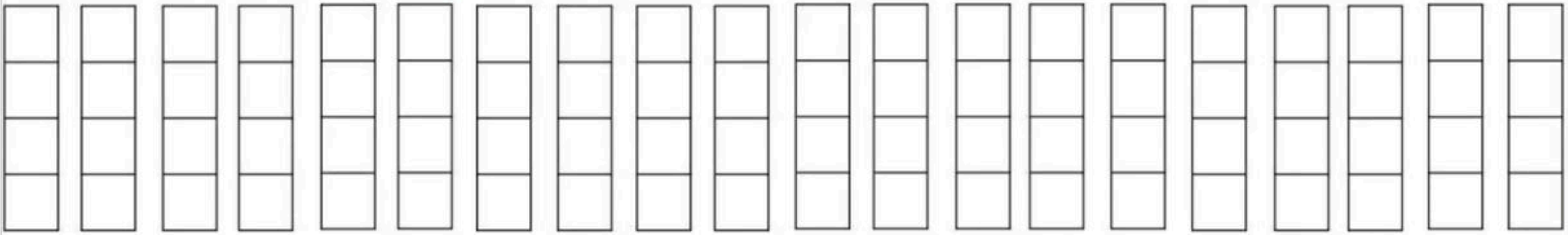
			2	2	2	2	2
		1	1	1	4	1	1
	0	0	0	0	0	0	0
7	7	7	7	3	3	3	7

Most Recently Used (MRU)

- The page which was used recently will be replaced first.

Example: Let the blocks be in the sequence: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 and the cache memory has 4 lines.

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1



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Example: Let the blocks be in the sequence: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 and the cache memory has 4 lines.

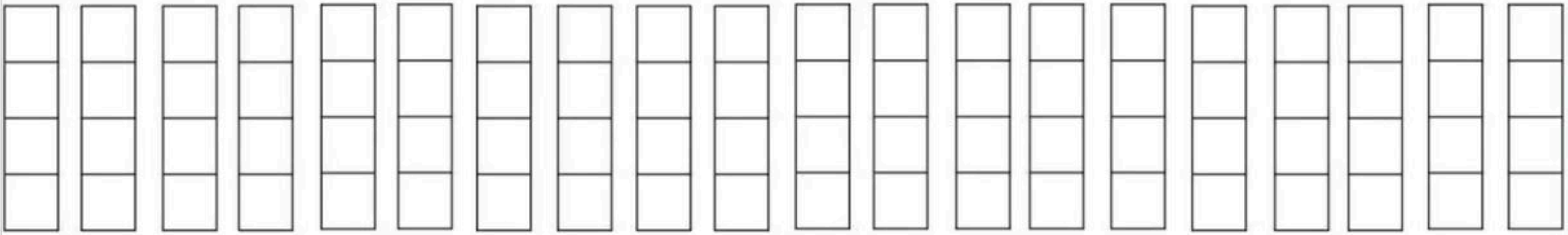
			2	2	2	2	3	0	3	2	0
		1	1	1	1	1	1	1	1	1	1
	0	0	0	3	0	4	4	4	4	4	4
7	7	7	7	7	7	7	7	7	7	7	7

Optimal Algorithm

- The page which will not be used for the longest period of time in future references will be replaced first.
- The optimal algorithm will provide the best performance but it is difficult to implement as it requires the future knowledge of pages which is not possible.
- It is used as a benchmark for cache replacement algorithms.

Example: Let the blocks be in the sequence: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 and the cache memory has 4 lines.

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1



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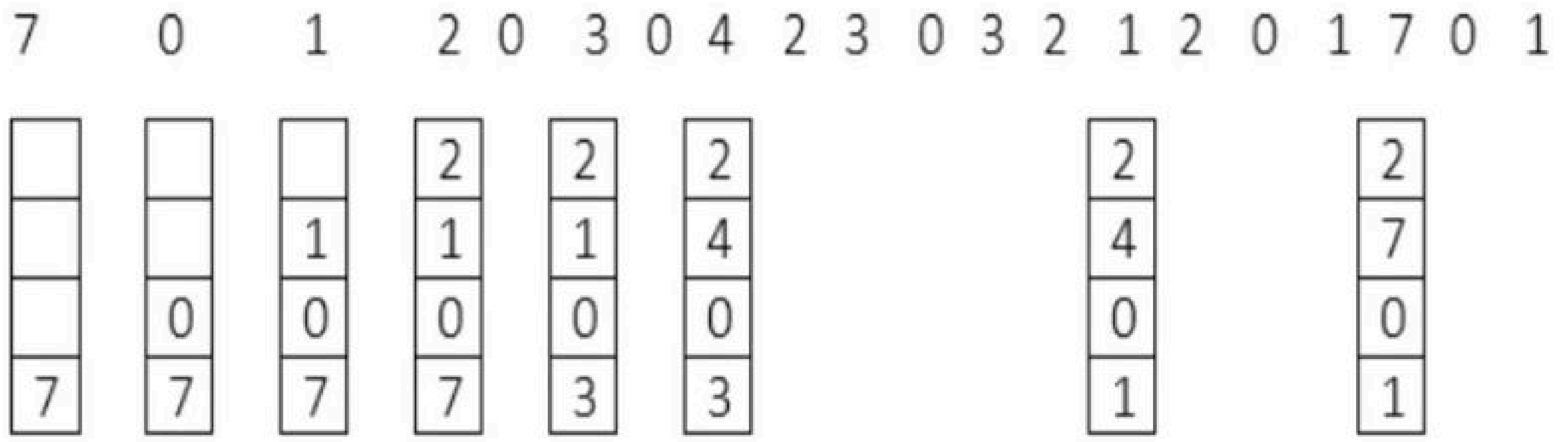
Example: Let the blocks be in the sequence: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 and the cache memory has 4 lines.



Break

Types of Miss

- **Compulsory Miss**
  - When CPU demands for any block for the first time then definitely a miss is going to occur as the block needs to be brought into the cache, it is known as Compulsory miss.

- **Capacity Miss**
  - Occur because blocks are being discarded from cache because cache cannot contain all blocks needed for program execution.

- **Conflict Miss**
  - In the case of set associative or direct mapped block placement strategies, conflict misses occur when several blocks are mapped to the same set or block frame; also called collision misses or interference misses.

Q Consider a 2-way set associative cache with 256 blocks and uses LRU replacement, Initially the cache is empty. Conflict misses are those misses which occur due the contention of multiple blocks for the same cache set. Compulsory misses occur due to first time access to the block. The following sequence of accesses to memory blocks
(0, 128, 256, 128, 0, 128, 256, 128, 1, 129, 257, 129, 1, 129, 257, 129) is repeated 10 times. The number of conflict misses experienced by the cache is ___________. **(Gate-2017) (2 Marks)**

Q Consider a 4-way set associative cache (initially empty) with total 16 cache blocks. The main memory consists of 256 blocks and the request for memory blocks is in the following order: 0, 255, 1, 4, 3, 8, 133, 159, 216, 129, 63, 8, 48, 32, 73, 92, 155. Which one of the following memory blocks will NOT be in cache if LRU replacement policy is used? **(Gate-2009) (2 Marks)**

(A) 3 **(B)** 8 **(C)** 129 **(D)** 216

Q Consider a Direct Mapped Cache with 8 cache blocks (numbered 0-7). If the memory block requests are in the following order 3, 5, 2, 8, 0, 63, 9,16, 20, 17, 25, 18, 30, 24, 2, 63, 5, 82,17, 24. Which of the following memory blocks will not be in the cache at the end of the sequence? **(GATE-2007) (2 Marks)**

(A) 3 **(B)** 18 **(C)** 20 **(D)** 30

Q Consider a Direct Mapped Cache with 8 cache blocks (numbered 0-7). If the memory block requests are in the following order 3, 5, 2, 8, 0, 63, 9,16, 20, 17, 25, 18, 30, 24, 2, 63, 5, 82,17, 24. Which of the following memory blocks will not be in the cache at the end of the sequence? **(GATE-2007) (2 Marks)**

(A) 3 **(B)** 18 **(C)** 20 **(D)** 30

3, 5, 2, 8, 0, 63, 9,16, 20, 17, 25, 18, 30, 24, 2, 63, 5, 82,17, 24

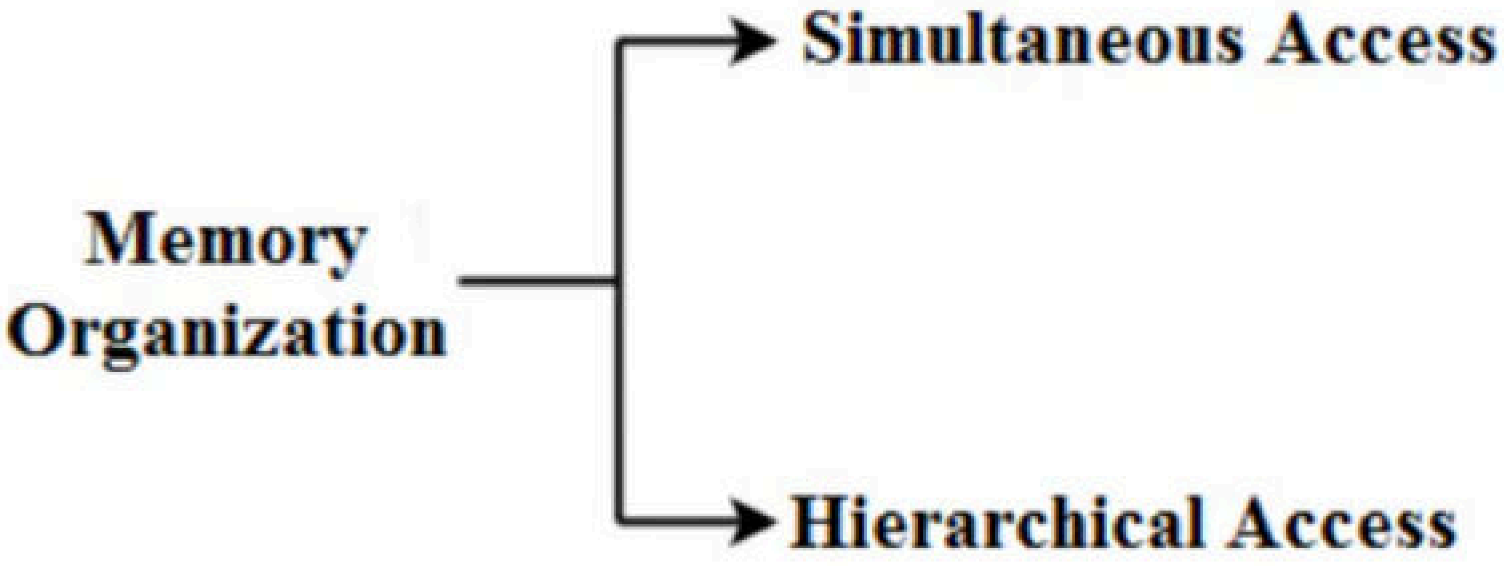
Q Consider a small two-way set-associative cache memory, consisting of four blocks. For choosing the block to be replaced, use the least recently used (LRU) scheme. The number of cache misses for the following sequence of block addresses is 8, 12, 0, 12, 8 **(Gate-2004) (2 Marks)**

(A) 2 **(B)** 3 **(C)** 4 **(D)** 5

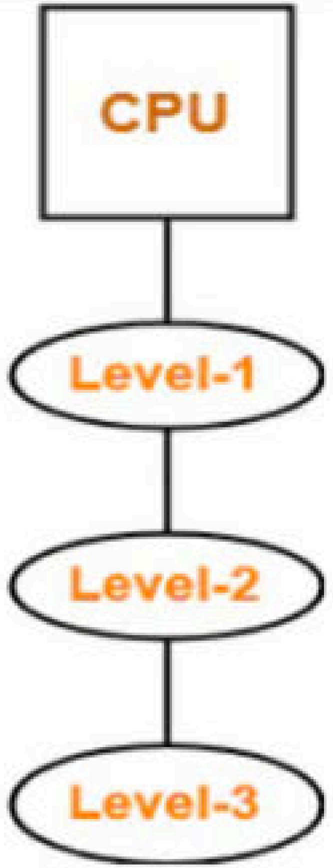
Break

Memory Organization

- Memory is organized at different levels.
- CPU may try to access different levels of memory in different ways.
- On this basis, the memory organization is broadly divided into two types



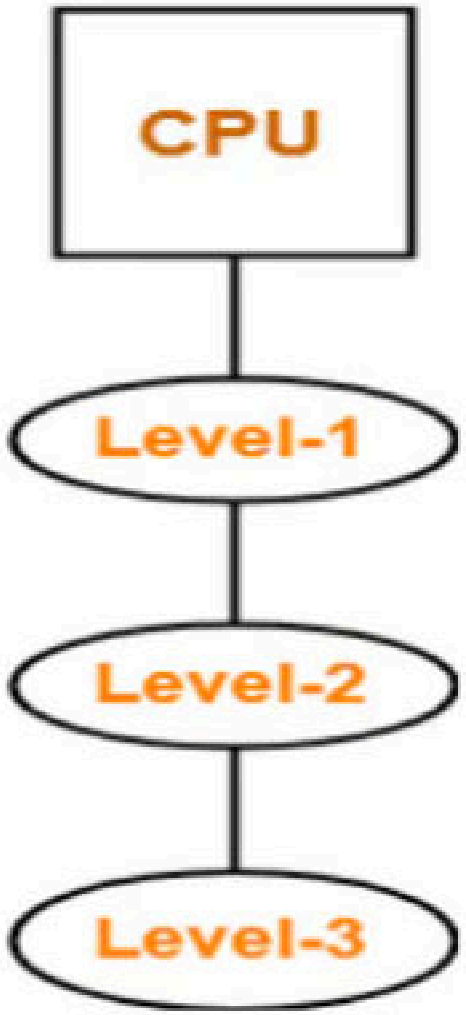
Hierarchical Access Memory Organization



- In this memory organization, memory levels are organized as
  - Level-1 is directly connected to the CPU.
  - Level-2 is directly connected to level-1.
  - Level-3 is directly connected to level-2 and so on

- Whenever CPU requires any word,
  - It first searches for the word in level-1.
  - If the required word is not found in level-1, it searches for the word in level-2.
  - If the required word is not found in level-2, it searches for the word in level-3 and so on.

• T_1 = Access time of level L_1 • S_1 = Size of level L_1 • C_1 = Cost per byte of level L_1 • H_1 = Hit rate of level L_1	• T_2 = Access time of level L_2 • S_2 = Size of level L_2 • C_2 = Cost per byte of level L_2 • H_2 = Hit rate of level L_2	• T_3 = Access time of level L_3 • S_3 = Size of level L_3 • C_3 = Cost per byte of level L_3 • H_3 = Hit rate of level L_3



Effective Memory Access Time

Average time required to access memory per operation =

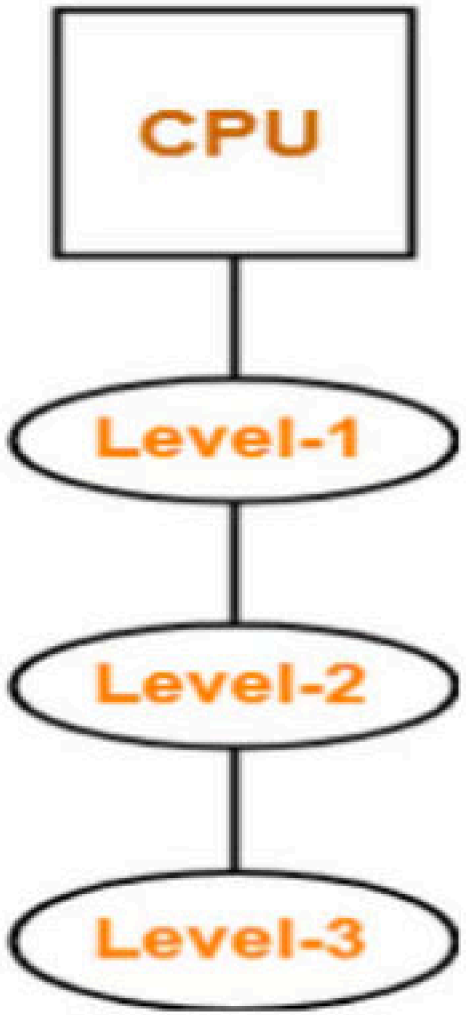
$$H_1 * T_1 + (1 - H_1) * H_2 * (T_1 + T_2) + (1 - H_1)(1 - H_2) * H_3 * (T_1 + T_2 + T_3)$$

$$H_1 * T_1 + (1 - H_1) * H_2 * (T_1 + T_2) + (1 - H_1)(1 - H_2) * (T_1 + T_2 + T_3)$$

Average Cost per Byte

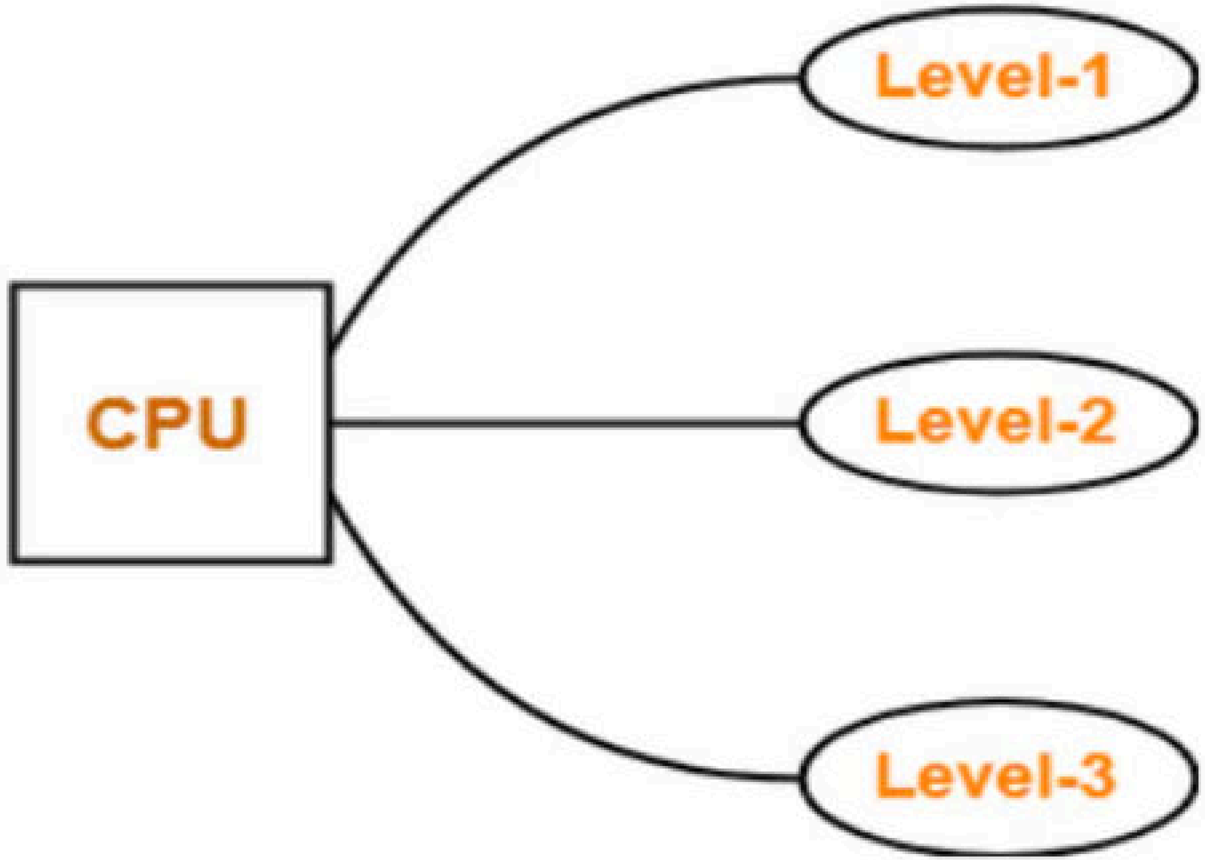
- Average cost per byte of the memory =

$$\frac{C_1 S_1 + C_2 S_2 + C_3 S_3}{S_1 + S_2 + S_3}$$



Simultaneous Access Memory Organization

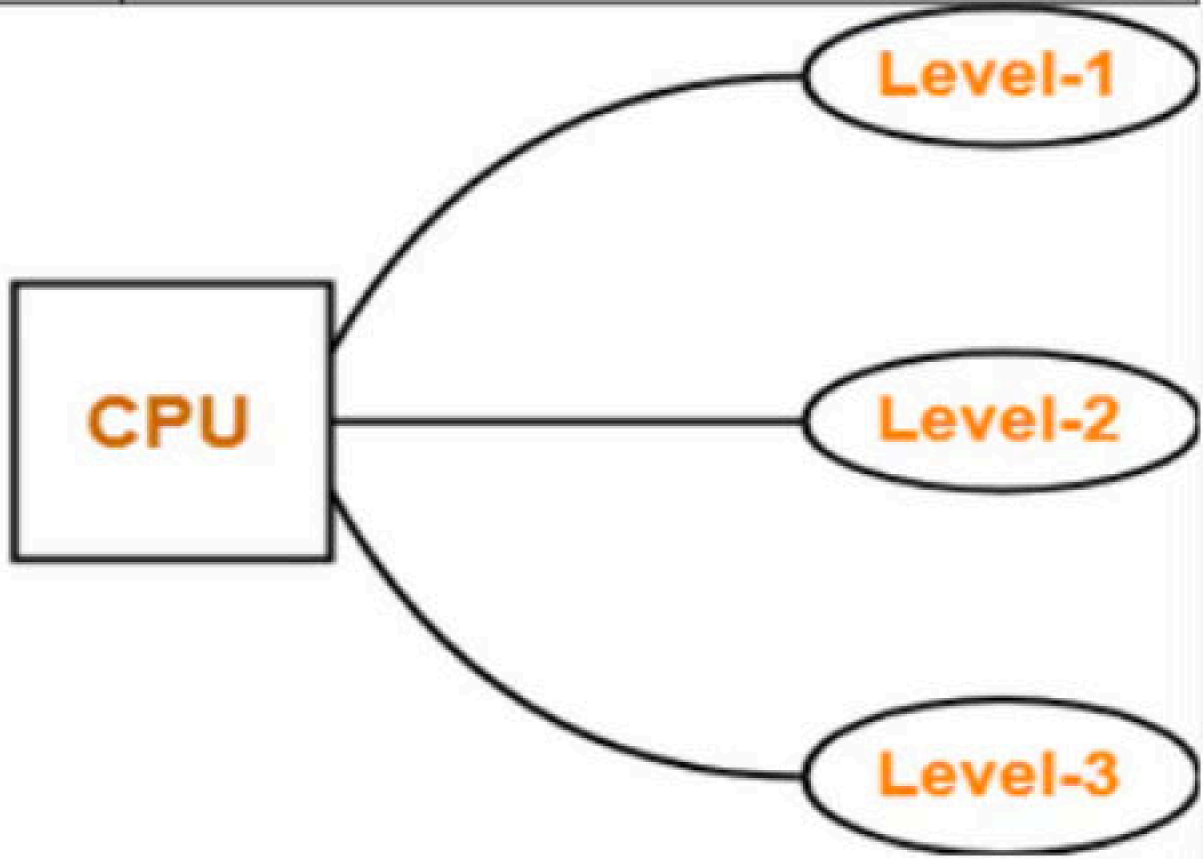
- In simultaneous access all the levels of memory are directly connected to the CPU, whenever CPU requires any word, it starts searching for it in all the levels simultaneously.



• T_1 = Access time of level L_1 • S_1 = Size of level L_1 • C_1 = Cost per byte of level L_1 • H_1 = Hit rate of level L_1	• T_2 = Access time of level L_2 • S_2 = Size of level L_2 • C_2 = Cost per byte of level L_2 • H_2 = Hit rate of level L_2	• T_3 = Access time of level L_3 • S_3 = Size of level L_3 • C_3 = Cost per byte of level L_3 • H_3 = Hit rate of level L_3

Effective Memory Access Time (EMAT) =

$$H_1 * T_1 + (1 - H_1) * H_2 * T_2 + (1 - H_1)(1 - H_2) * H_3 * T_3$$



In any memory organization, the data item being searched will definitely be present in the last level (or secondary memory).

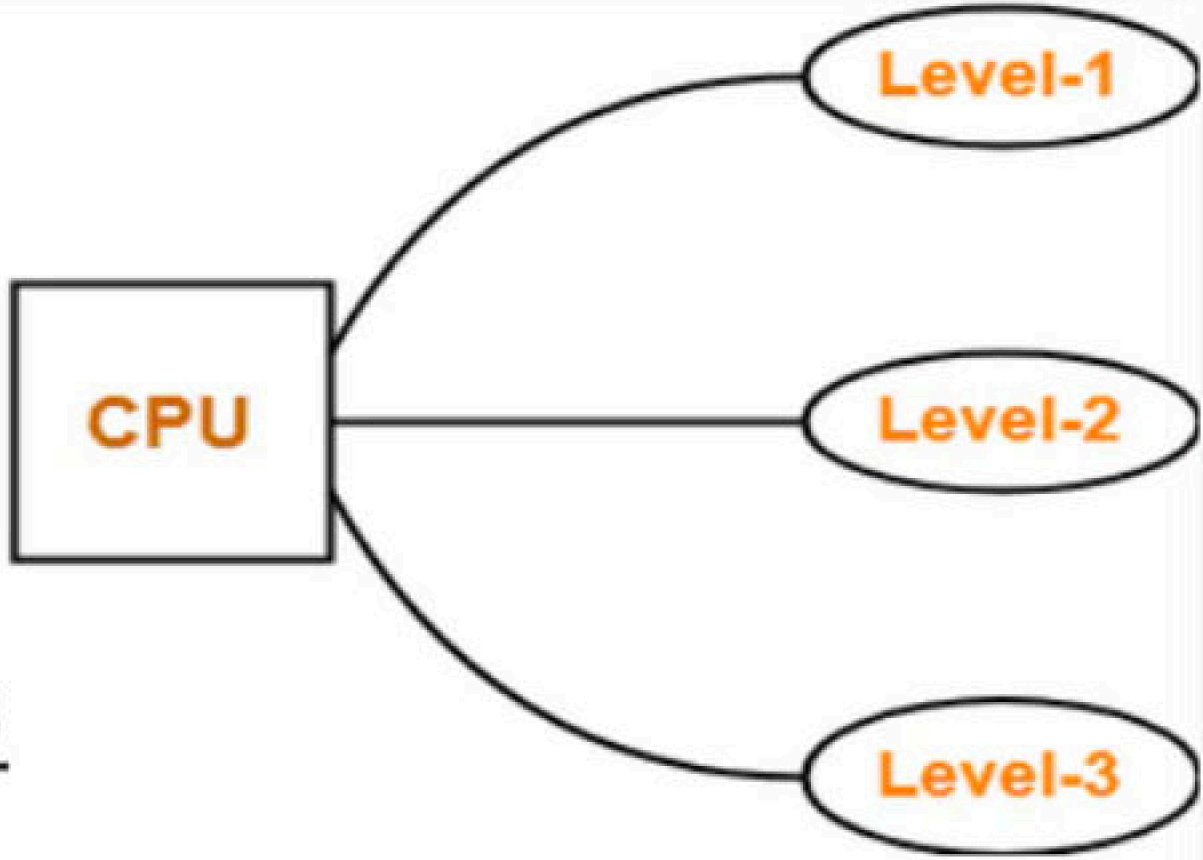
Thus, hit rate for the last level is always 1. So,

$$H_1 * T_1 + (1 - H_1) * H_2 * T_2 + (1 - H_1)(1 - H_2) * T_3$$

Average Cost per Byte

- Average cost per byte of the memory =

$$\frac{C_1 S_1 + C_2 S_2 + C_3\, S_3}{S_1 + S_2 + S_3}$$



Example: Calculate the EMAT for a machine with a cache hit rate of 80% where cache access time is 5ns and main memory access time is 100ns, both for simultaneous and hierarchical access.

Q Assume that for a certain processor, a read request takes 50 nanoseconds on a cache miss and 5 nanoseconds on a cache hit. Suppose while running a program, it was observed that 80% of the processor's read requests result in a cache hit. The average read access time in nanoseconds is_____________. **(GATE-2015) (2 Marks)**

Q Consider a system with 2 level cache. Access times of Level 1 cache, Level 2 cache and main memory are 0.5 ns, 5 ns and 100 ns respectively. The hit rates of Level 1 and Level 2 caches are 0.7 and 0.8 respectively. What is the average access time of the system ignoring the search time within the cache? **(NET-DEC-2018)**

a) 35.20 ns **b)** 7.55 ns **c)** 20.75 ns **d)** 24.35 ns

Cache Coherence Problem

- If multiple copy of same data is maintained at different level of memories then inconsistency may occur, this problem is known as cache coherence problem.

- Cache coherence problem can be resolved using the following techniques:
  - Write Through
  - Write Back

Write Through

- Write through is used to maintain the consistency between the cache and main memory.
- According to it if the cache copy is updated, at the same time main memory is also updated.

- **Advantages**
  - It provides the highest level of consistency.

- **Disadvantages**
  - It requires more number of memory access.

Write Back

- Write back is also used to maintain the consistency between the cache and main memory.
- According to it all the changes performed on cache are reflected back to the main memory in the end.

- **Advantage**
  - Less number of memory accesses and less write operations.

- **Disadvantage**
  - Inconsistency may occur.

Q In ______ method, the word is written to the block in both the cache and main memory, in parallel. **(NET-JULY-2016)**

(a) Write through – 84

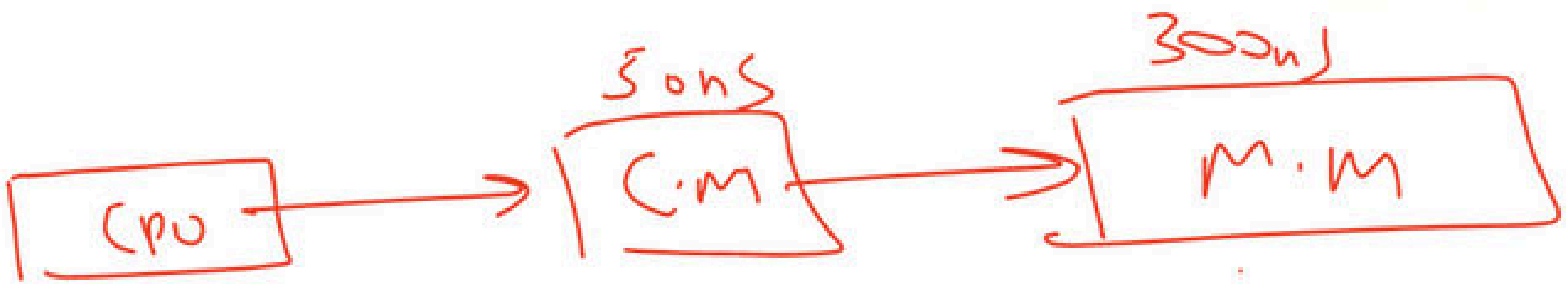
(b) Write back – 14

(c) ~~Write protected~~ – 0

(d) ~~Direct mapping~~ – 1

Q A hierarchical memory system that uses cache memory has cache access time of 50 nano seconds, main memory access time of 300 nano seconds, 75% of memory requests are for read, hit ratio of 0.8 for read access and the write-through scheme is used. What will be the average access time of the system both for read and write requests? **(NET-DEC-2014)** 40

(A) 157.5 n.sec **(B)** 110 n.sec 32 **C)** 75 n.sec 15 **(D)** 82.5 n.sec 13



$T_R = \cancel{.8 \times 50ns} + (1 - .8)[50 + 300]$

$40ns + .2 \times 350 = \underline{110nS}$

$T_W = \max(50, 300) = 300nS$

$T = \underline{.75 \times 110nS} + \underline{.25 \times 300nS} = \underline{157.5}$

Q For inclusion to hold between two cache levels L_1 and L_2 in a multi-level cache hierarchy, which of the following are necessary? **(Gate-2008) (2 Marks)**

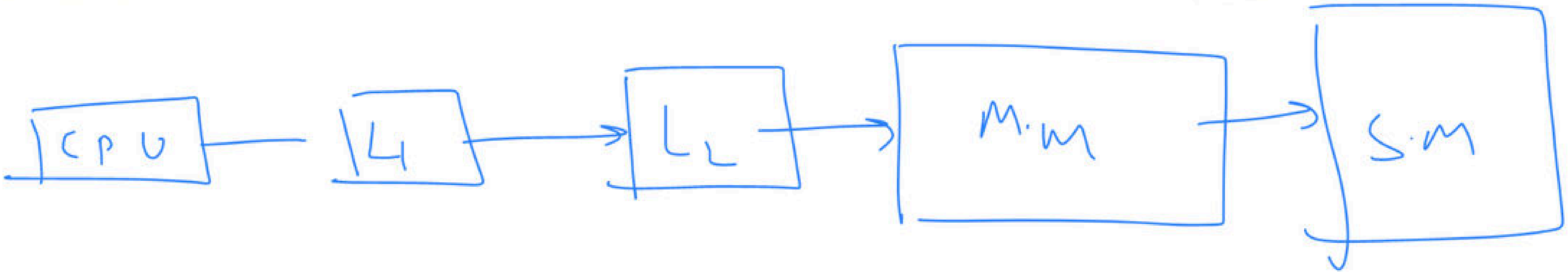
I. L_1 must be a write-through cache

II. L_2 must be a write-through cache

III. The associativity of L_2 must be greater than that of L1

IV. The L_2 cache must be at least as large as the L1 cache

(A) IV only — 40
(B) I and IV only — 19
(C) I, III and IV only — 27
(D) I, II, III and IV — 13



$L_1 \subseteq L_2 \subseteq M.M \subseteq S.M$

Q In a two-level cache system, the access times of L_1 and L_2 is 1 and 8 clock cycles, respectively. The miss penalty from the L_2 cache to main memory is 18 clock cycles. The miss rate of L_1 cache is twice that of L_2. The average memory access time (AMAT) of this cache system is 2 cycles. The miss rates of L_1 and L_2 respectively are: **(Gate-2017) (2 Marks)**

(A) 0.111 and 0.056 — 43
(B) 0.056 and 0.111 — 27
(C) 0.0892 and 0.1784 — 18
(D) 0.1784 and 0.0892 — 11

$$\text{CPU} \rightarrow \overset{1}{L_1} \rightarrow \overset{8}{L_2} \rightarrow \overset{18}{\text{M.M}}$$

L_1: M $2x$, H $1-2x$
L_2: M x, H $1-x$

$$2 = (1-2x)[1] + 2x\cdot(1-x)[1+8] + 2x\cdot x[1+8+18]$$

$$2 = 1-2x + 18x(1-x) + 54x^2$$

$$2 = 1 - 2x + 18x - 18x^2 + 54x^2$$

$$36x^2 + 16x - 1 = 0$$

$$\Rightarrow 36x^2 + 18x - 2x - 1 = 0$$

$$18x(2x+1) - 1(2x+1) = 0$$

$$(18x-1)(2x+1) = 0$$

$$x = \frac{1}{18}$$

$18x^2 + 16x - 1 = 0$

$18x^2 + 18x - 2x - 1 = 0$

$\frac{1+8+18}{27}$

$x = \frac{1}{18} \;\Big|\; \cancel{x = -1}$

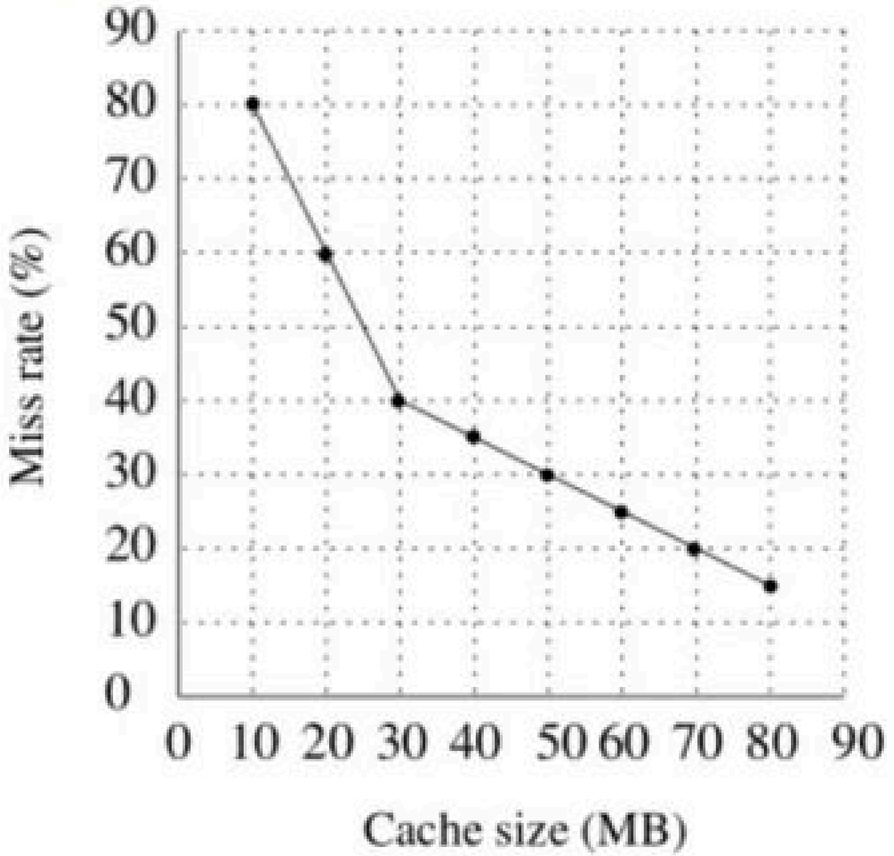
Q The read access times and the hit ratios for different caches in a memory hierarchy are as given below:

Cache	Read access time (in nanoseconds)	Hit ratio
I-cache	2	0.8
D-cache	2	0.9
L2-cache	8	0.9

The read access time of main memory in 90 nanoseconds. Assume that the caches use the referred-word-first read policy and the writeback policy. Assume that all the caches are direct mapped caches. Assume that the dirty bit is always 0 for all the blocks in the caches. In execution of a program, 60% of memory reads are for instruction fetch and 40% are for memory operand fetch. The average read access time in nanoseconds (up to 2 decimal places) is __________ **(Gate-2017) (2 Marks)**

Q Consider a two-level cache hierarchy L_1 and L_2 caches. An application incurs 1.4 memory accesses per instruction on average. For this application, the miss rate of L_1 cache 0.1, the L_2 cache experience on average. 7 misses per 1000 instructions. The miss rate of L_2 expressed correct to two decimal places is _______________. **(Gate-2017) (1 Marks)**

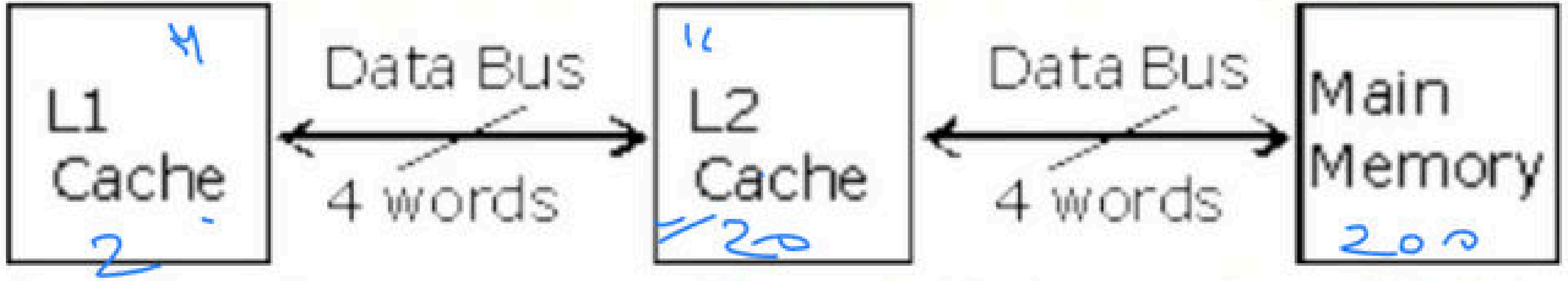
Q A file system uses an in-memory cache to cache disk blocks. The miss rate of the cache is shown in the figure. The latency to read a block from the cache is 1 ms and to read a block from the disk is 10 ms. Assume that the cost of checking whether a block exists in the cache is negligible. Available cache sizes are in multiples of 10 MB.



The smallest cache size required to ensure an average read latency of less than 6 ms is _______ MB. **(Gate-2016) (2 Marks)**

Q The memory access time is 1 nanosecond for a read operation with a hit in cache, 5 nanoseconds for a read operation with a miss in cache, 2 nanoseconds for a write operation with a hit in cache and 10 nanoseconds for a write operation with a miss in cache. Execution of a sequence of instructions involves 100 instruction fetch operations, 60 memory operand read operations and 40 memory operands write operations. The cache hit-ratio is 0.9. The average memory access time (in nanoseconds) in executing the sequence of instructions is __________. **(Gate-2014) (2 Marks)**

Q A computer system has an L_1 cache, an L_2 cache, and a main memory unit connected as shown below. The block size in L_1 cache is 4 words. The block size in L_2 cache is 16 words. The memory access times are 2 nanoseconds. 20 nanoseconds and 200 nanoseconds for L_1 cache, L_2 cache and main memory unit respectively.



When there is a miss in L_1 cache and a hit in L_2 cache, a block is transferred from L_2 cache to L_1 cache. What is the time taken for this transfer? **(Gate-2010) (2 Marks)**

(A) 2 nanoseconds — 3

(B) 20 nanoseconds — 17

(C) 22 nanoseconds — 76

(D) 88 nanoseconds — 10

L1L2

18

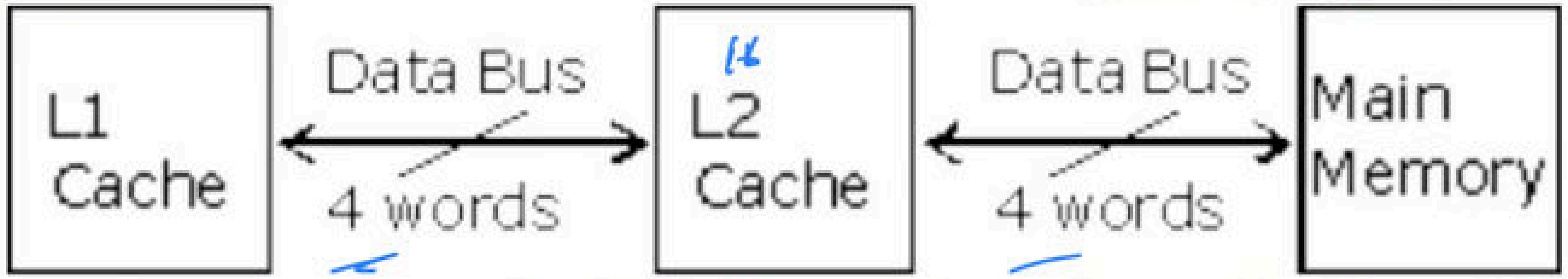
ch

/32

2+20

22

Q A computer system has an L_1 cache, an L_2 cache, and a main memory unit connected as shown below. The block size in L_1 cache is 4 words. The block size in L_2 cache is 16 words. The memory access times are 2 nanoseconds. 20 nanoseconds and 200 nanoseconds for L_1 cache, L_2 cache and main memory unit respectively.



Q When there is a miss in both L_1 cache and L_2 cache, first a block is transferred from main memory to L_2 cache, and then a block is transferred from L_2 cache to L_1 cache. What is the total time taken for these transfers? **(Gate-2010) (2 Marks)**

(A) 222 nanoseconds — 39 **(B)** 888 nanoseconds — 13
(C) 902 nanoseconds — 43 **(D)** 968 nanoseconds — 4

$$\boxed{(200+20)\times 4} + 20+2$$

~~220x4~~

$880 + 20 + 2 = \underline{902}$

Q Consider a system with 2 level caches. Access times of $Level_1$ cache, $Level_2$ cache and main memory are 1 ns, 10ns, and 500 ns, respectively. The hit rates of $Level_1$ and $Level_2$ caches are 0.8 and 0.9, respectively. What is the average access time of the system ignoring the search time within the cache? **(Gate-2004) (1 Marks)**

(A) 13.0 ns 7
(B) 12.8 ns 26
(C) 12.6 ns 73
(D) 12.4 ns 0

$$\text{CPU} \rightarrow \underset{0.8}{\overset{1nS}{L_1}} \rightarrow \underset{0.9}{\overset{10nS}{L_2}} \rightarrow \overset{500nS}{M.M}$$

$$= .8[1nS] + .9(1-.8)[10nS] + (1-.8)(1-.9)1(500)$$

$$= \underline{\underline{12.6nS}}$$

Q A dynamic RAM has a memory cycle time of 64 nsec. It has to be refreshed 100 times per msec and each refresh takes 100 nsec. What percentage of the memory cycle time is used for refreshing? **(Gate-2005) (1 Marks)**

(A) 10 **(B)** 6.4 **(C)** 1 **(D)** .64

Q A cache line is 64 bytes. The main memory has latency 32ns and bandwidth 1G.Bytes/s. The time required to fetch the entire cache line from the main memory is **(Gate-2006) (2 Marks)**

(A) 32 ns **(B)** 64 ns **(C)** 96 ns **(D)** 128 ns

Q.29 Assume a two-level inclusive cache hierarchy. L1 and L2, where L2 is the larger of the two. Consider the following statements.

S_1 : Read misses in a write through L1 cache do not result in writebacks of dirty lines to the L2.

S_2 : Write allocate policy must be used in conjunction with write through caches and no-write allocate policy is "used with writeback caches.

(GATE- 2021)

Which of the following statements is correct?

(a) S_1 is true and S_2 is true.
(b) S_1 is true and S_2 is false.
(c) S_1 is false and S_2 is true.
(d) S_1 is false and S_2 is false.

Input / Output management

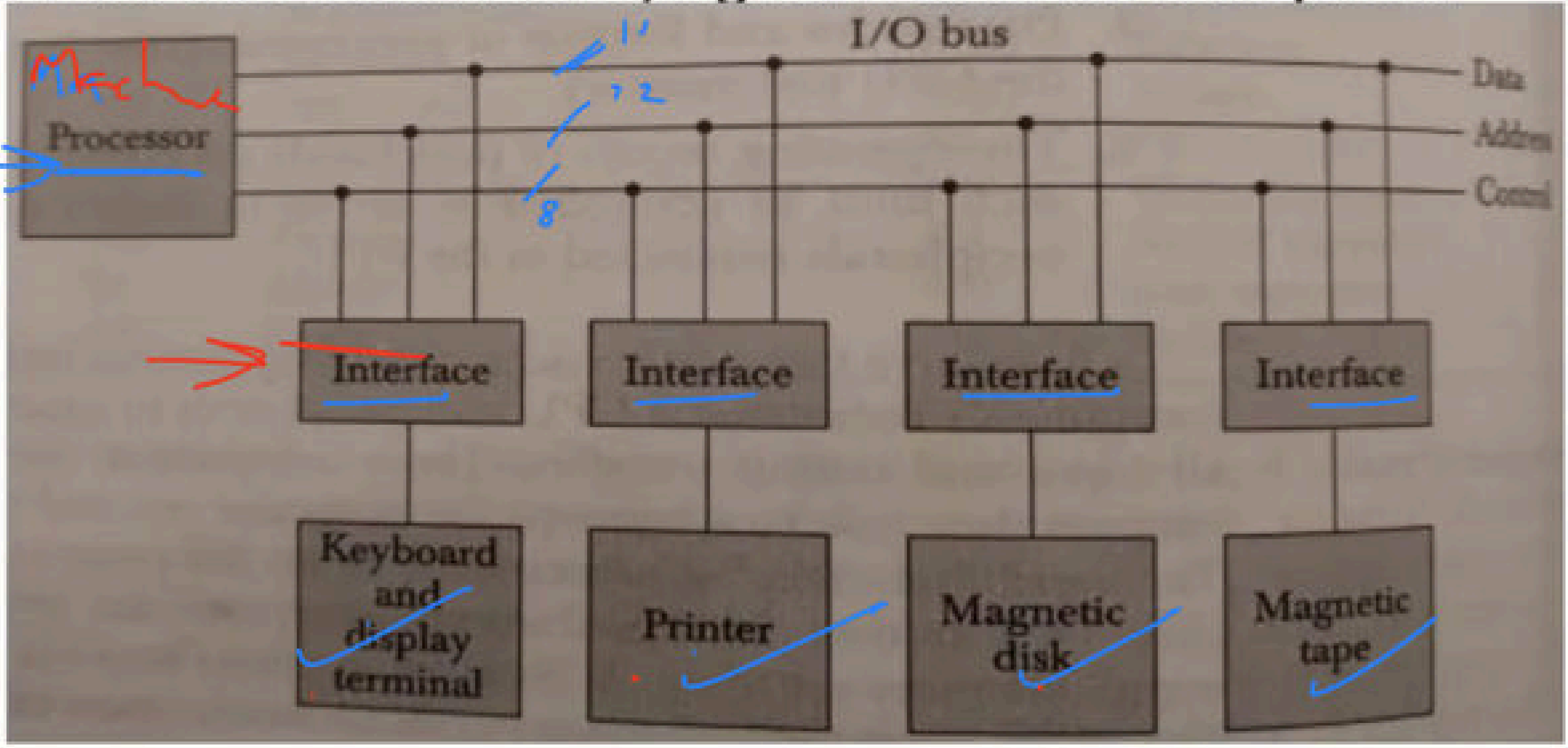
- In general, when we say a computer, we understand a CPU + Memory (cache, main memory)
- But a computer does not serve any purpose if it cannot receive data from the outside world or cannot transmit the data to outside world.
- I/o or peripheral devices are those independent devices which serve this purpose.
- So, while designing i/o for a computer we must know the number of i/o device and the capacity of each device.

Interface

CPHM

we cannot directly connect a i/o device to computer because of the following reasons

- **Speed: -** The speed of CPU and i/o device will usually different.
- **Format: -** The data code and format of CPU and peripherals may be different. E.g. ASCII, Unicode etc.
- **Physical orientation: -** Different device have organizations like optical, magnetic, electrochemical and different controlling functions.
- **Signal conversion: -** peripherals are electromagnetic and electrochemical device and their manner of operation is different from the operations of CPU and memory, which are electronic device, signal conversion is required

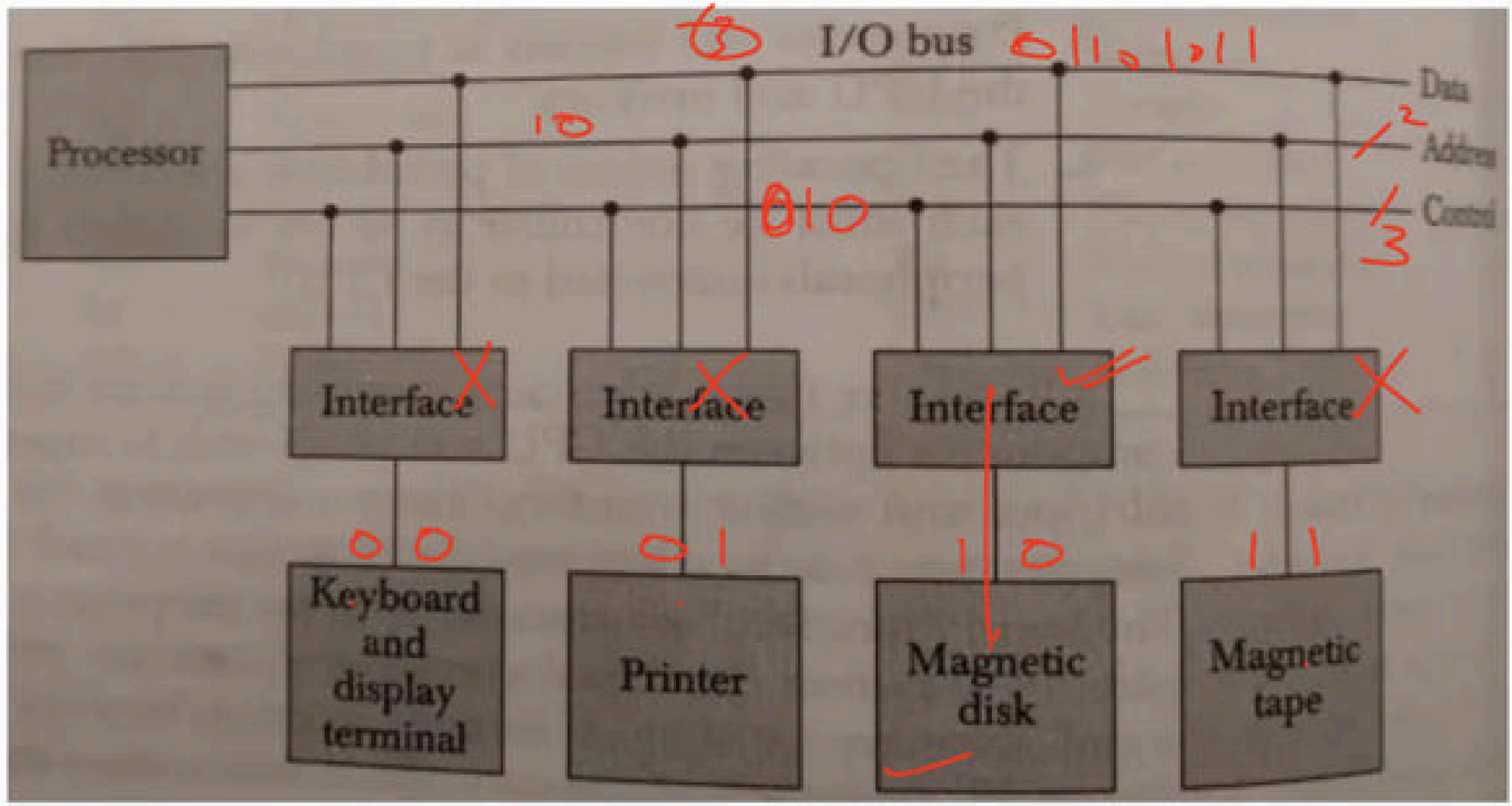


- **Address bus: -** Is used to identify the correct i/o devices among the number of i/o device , so CPU put an address of a specific i/o device on the address line, all devices keep monitoring this address bus and decode it, and if it is a match then it activates control and data lines.

- **Control bus: -** After selecting a specific i/o device CPU sends a functional code on the control line. The selected device(interface) reads that functional code and execute it. E.g. i/o command, control command, status command etc.

- **Data bus: -** In the final step depending on the operation either CPU will put data on the data line and device will store it or device will put data on the data line and CPU will store it.

R 000
W 001
Search 010
INFO 011
Format 100



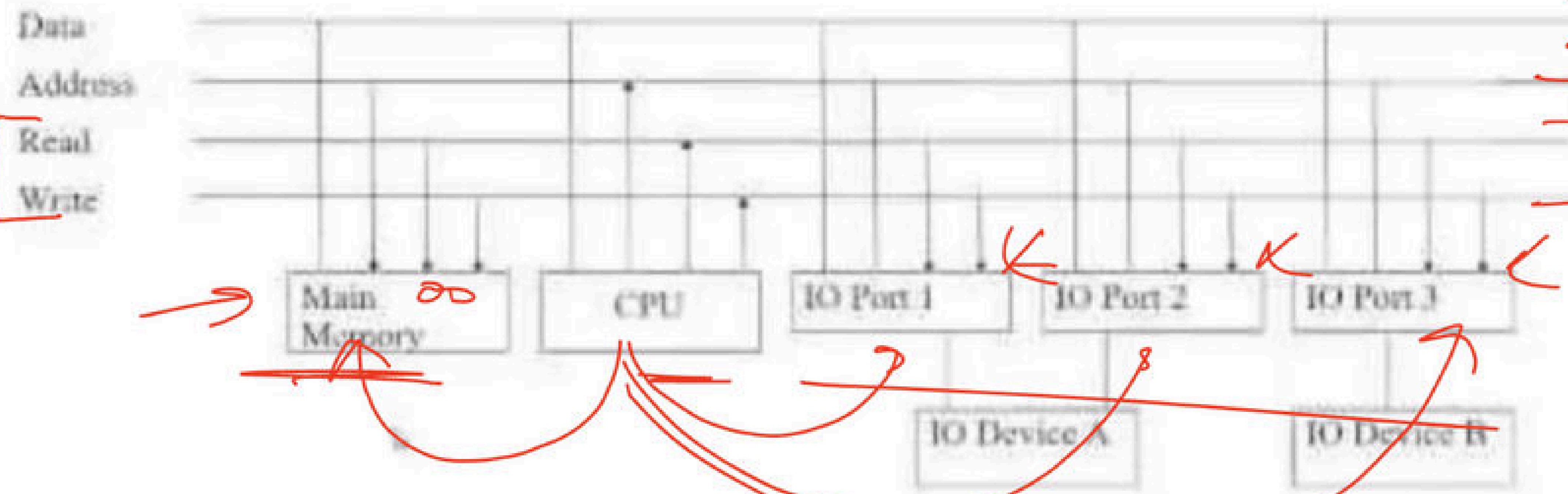
1990 h/w 2020

Q Consider a main memory system that consists of 8 memory modules attached to the system bus, which is one word wide. When a write request is made, the bus is occupied for 100 nanoseconds (ns) by the data, address, and control signals. During the same 100 ns, and for 500 ns thereafter, the addressed memory module executes one cycle accepting and storing the data. The (internal) operation of different memory modules may overlap in time, but only one request can be on the bus at any time. The maximum number of stores (of one word each) that can be initiated in 1 millisecond is ____________ **(Gate-2014) (2 Marks)**

(A) 1000 **(B)** 10000 **(C)** 100000 **(D)** 100

How a computer (CPU) deals with Memory and I/O devices

Memory Mapped i/o



- Here there is no separate i/o instructions. The CPU can manipulate i/o data residing in the interface registers with the same instructions that are used to manipulate memory words.
- Here computer can use memory type instructions for i/o data.
- E.g. 8085

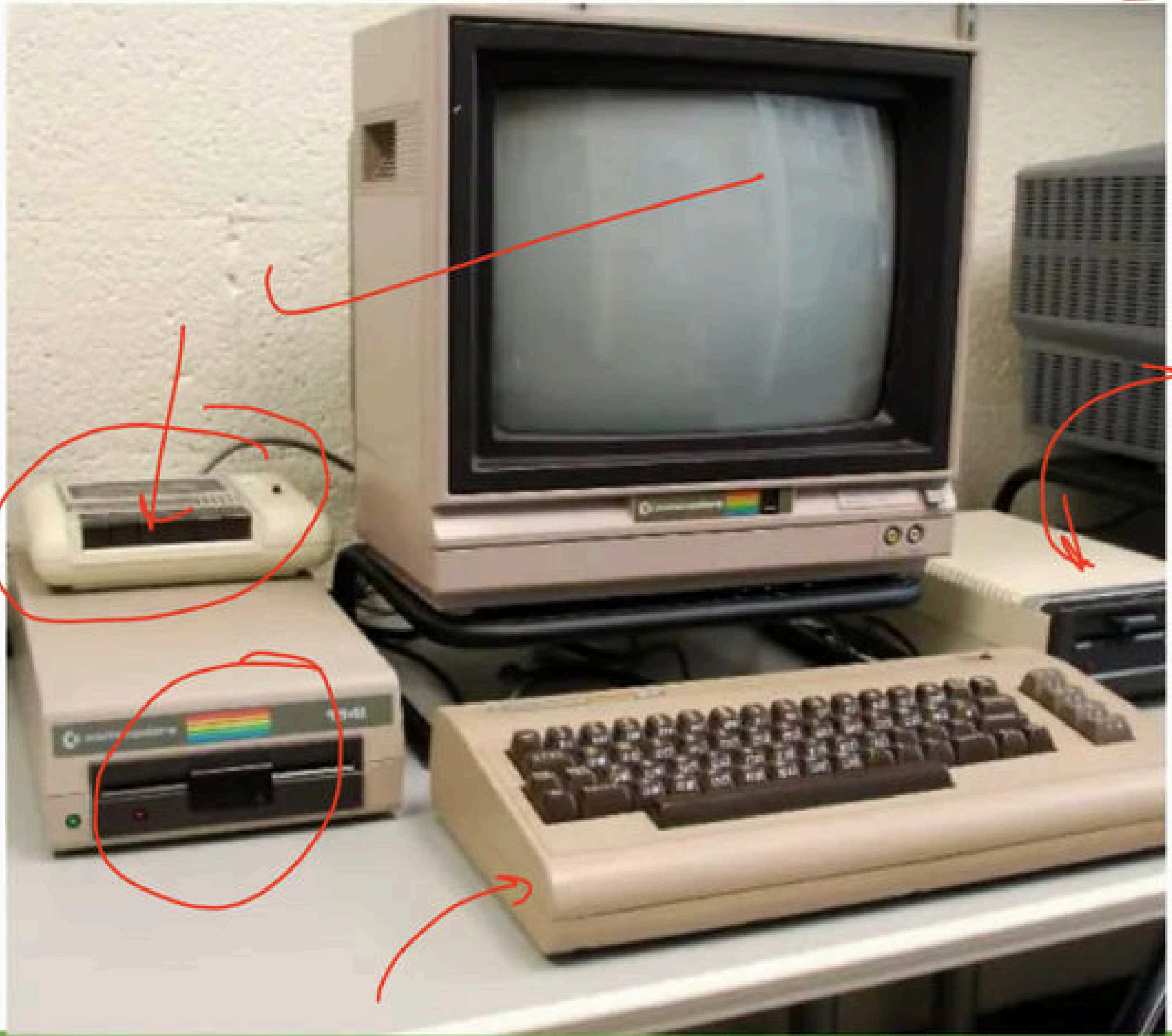
- **Advantage: -** In typical computer there are more memory reference instruction than i/o instruction, but in memory mapped i/o all instructions that refer to memory are also available for i/o.

- **Disadvantage: -**Total address get divided, some range is occupied by i/o while some memory.

- The **Commodore 64**, also known as the **C64** or the **CBM 64**, is an 8-bit home computer introduced in January 1982 — 40 by Commodore International (first shown at the Consumer Electronics Show, in Las Vegas, January 7–10, 1982).

- It has been listed in the Guinness World Records as the highest-selling single computer model of all time, with independent estimates placing the number sold between 10 and 17 million units.

1.5



Isolated i/o

- Here common bus to transfer data between memory or i/o and CPU. The distinction between a memory and i/o transfer is made through separate read and write line.
- i/o read and i/o write control lines are enabled during an i/o transfer.
- Memory read and memory write control lines are enabled during a memory transfer.

- **Advantage: -** Here memory is used efficiently as the same address can be used two times. E.g 8086

- **Disadvantage: -** Need different control lines one for memory and other for i/o devices.

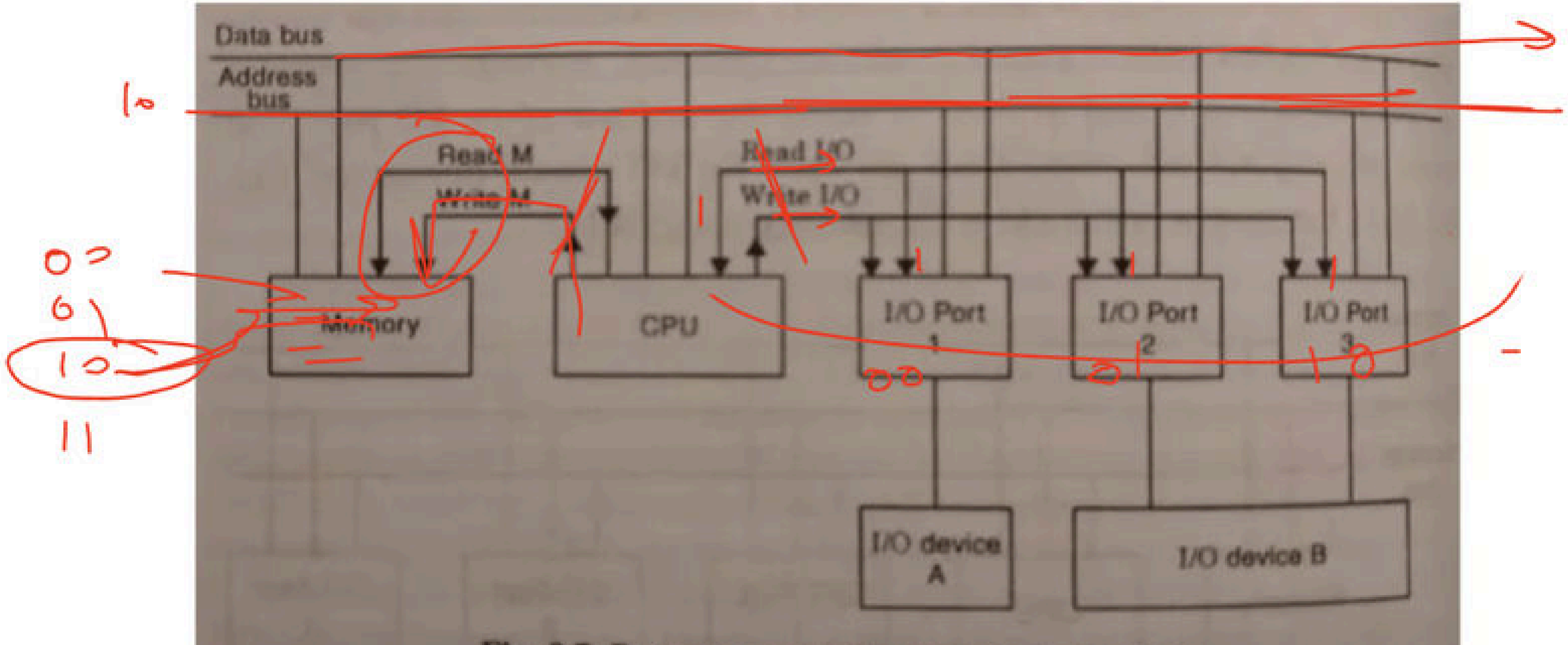


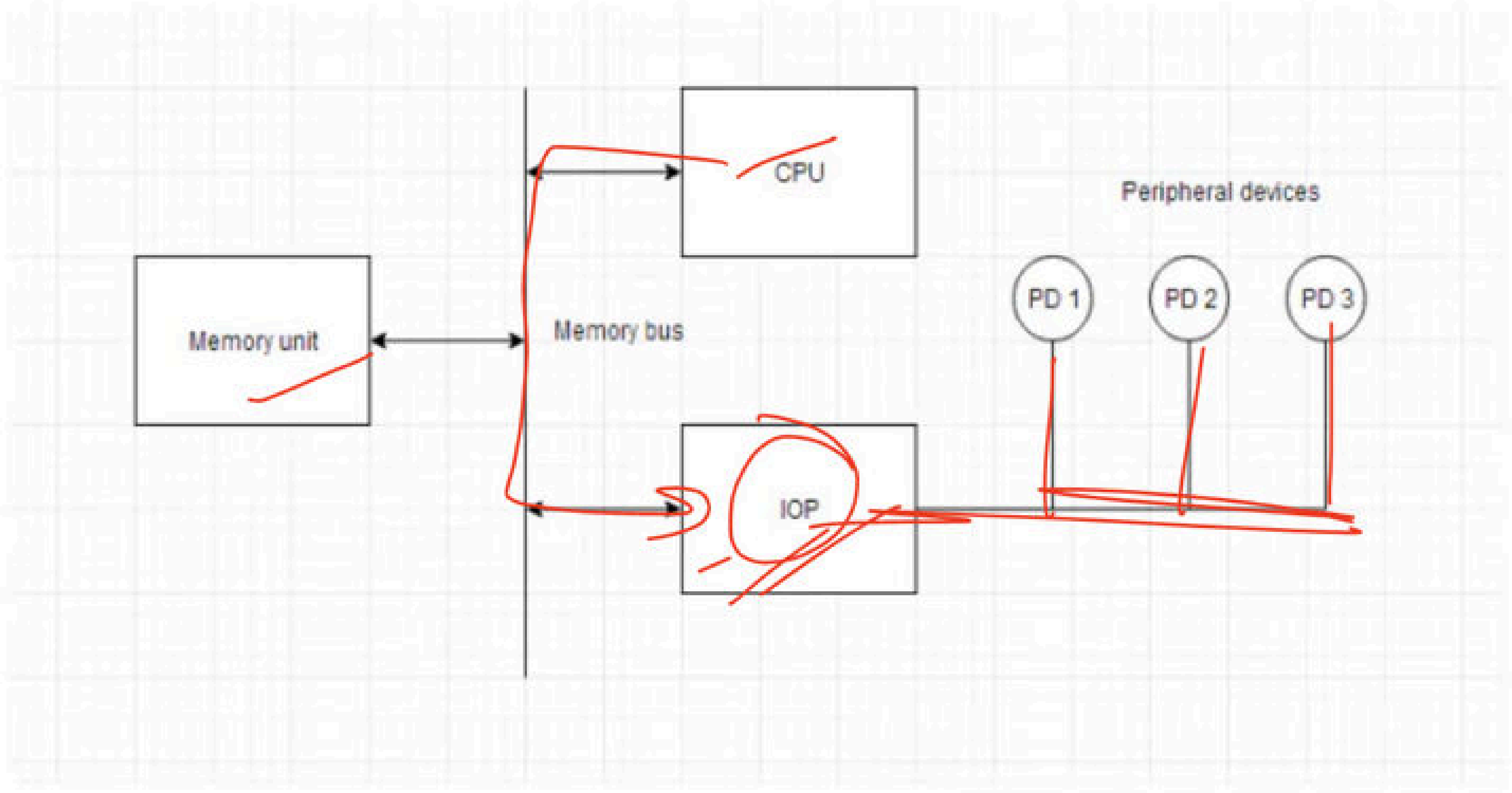
Fig. 8.7. Programmed I/O with I/O mapped I/O.

I/O Processor

- Computer has independent sets of data, address and control buses, one for accessing memory and the other for i/o. this is done in computers that provide a separate i/o processor other than CPU.
- Memory communicate with both the CPU and iop through a memory bus.
- Iop communicates also with the input and output devices through a separate i/o bus with its own address, data and control lines. The purpose of iop is to provide an independent pathway for the transfer of information between external devices and internal memory.

G, C

C, C



Synchronous Vs Asynchronous data transfer

- Synchronization is achieved by a device called master generator, which generate a periodic train of clock pulse.
- The internal operations in a digital system are synchronized by means of clock pulses supplied by a common pulse generator.
- When communication happens between devices which are under a same control unit or same clock then it is called synchronous communication. e.g. communication between CPU and its registers.

Asynchronous communication

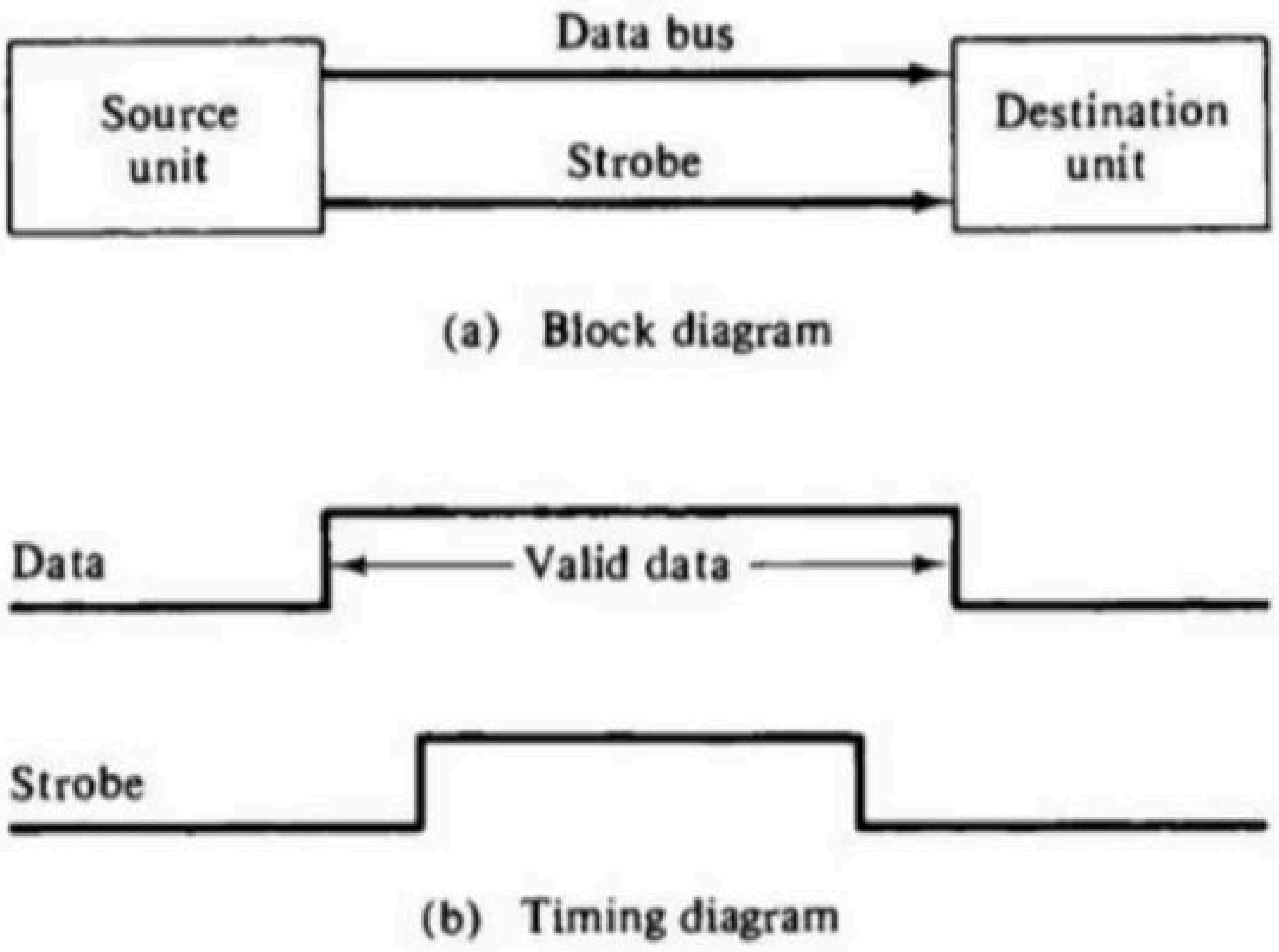
- When the timing units of two devices are independent that is they are under different control then it is called asynchronous communication.

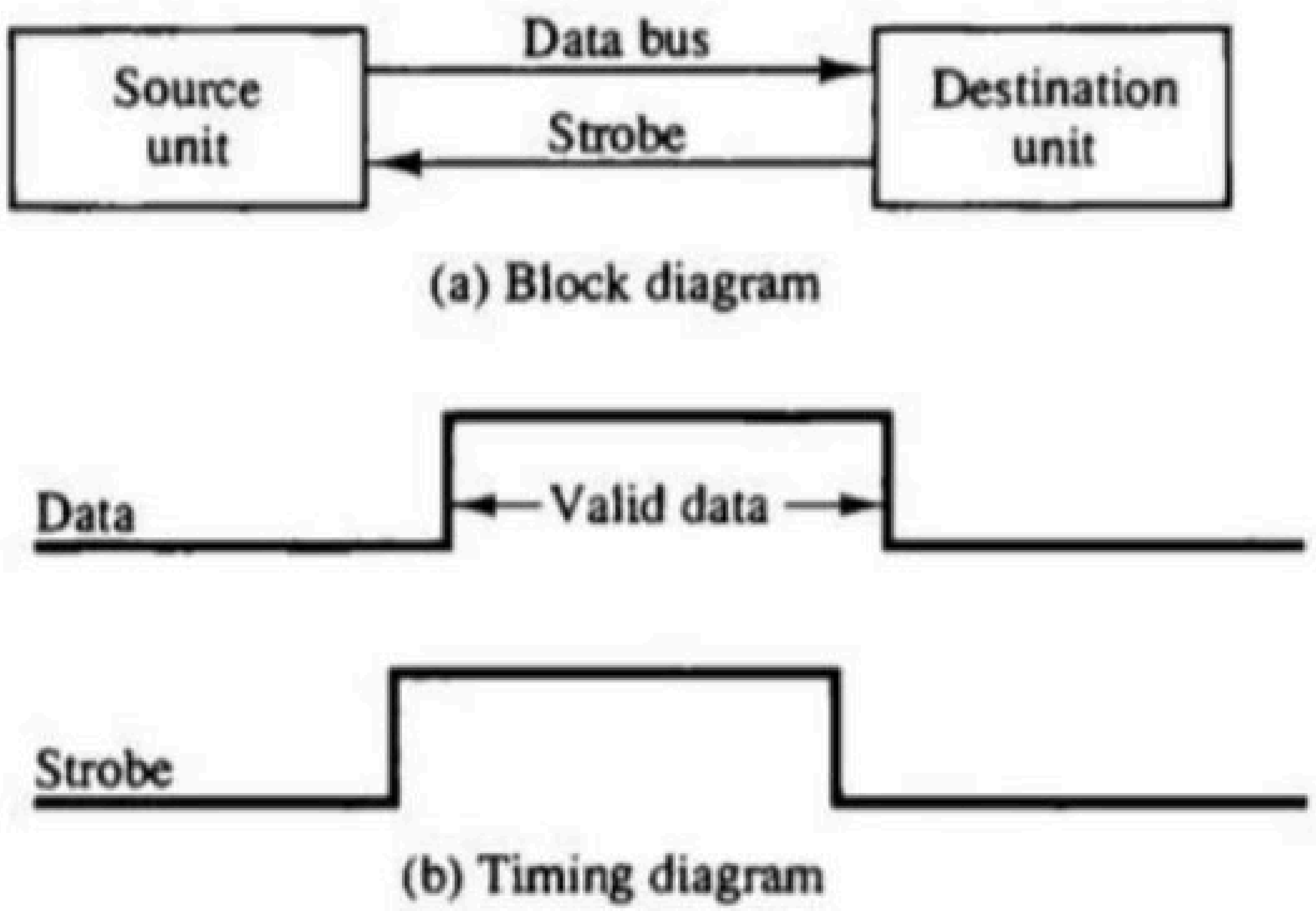
- Asynchronous data transfer between two independent units required that control signals be transmitted between the communicating units to indicate the time at which data is being transmitted.

- One way to achieving this by means of strobe pulse supplied by one of the units to indicate to the other unit which when the transfer has to occur.

- the unit receiving the data item response with another control signal to acknowledge the receipt of the data. this type of agreement between two independent units is referred to as handshaking.

- The strobe pulse method and the handshaking method of asynchronous data transfer are not restricted to input output transfer. in fact they are used extensively on numerous occasions requiring the transfer of data between two independent units.

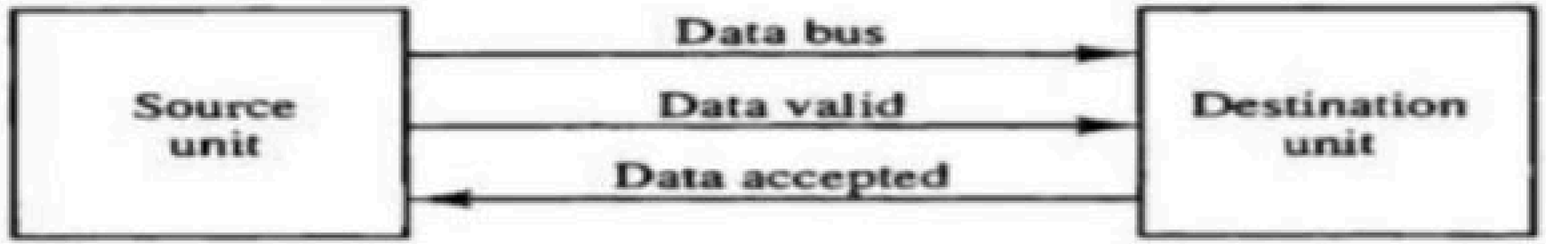




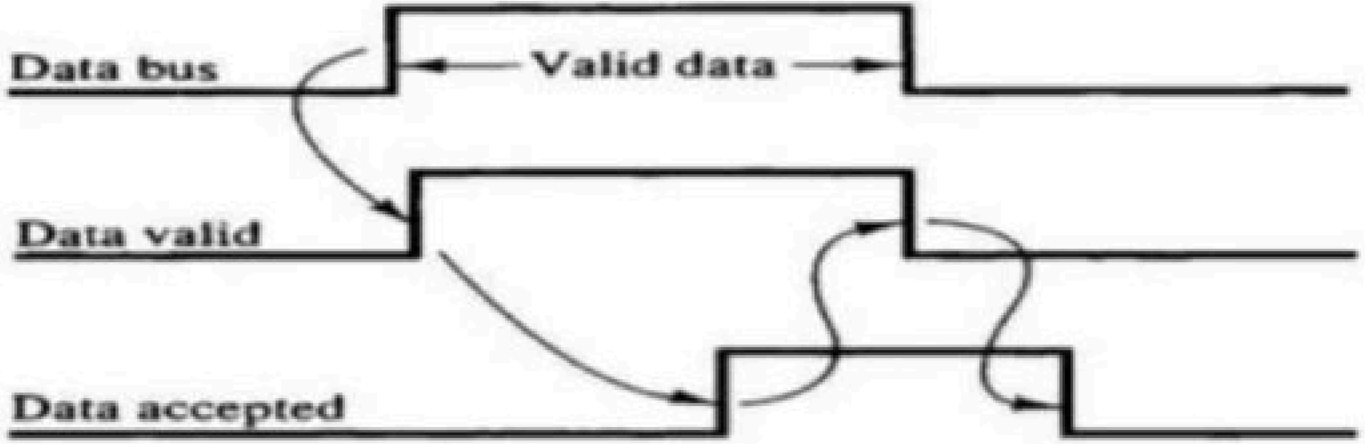
- the unit receiving the data item response with another control signal to acknowledge the receipt of the data. this type of agreement between two independent units is referred to as handshaking.

- The strobe pulse method and the handshaking method of asynchronous data transfer are not restricted to input output transfer. in fact they are used extensively on numerous occasions requiring the transfer of data between two independent units.

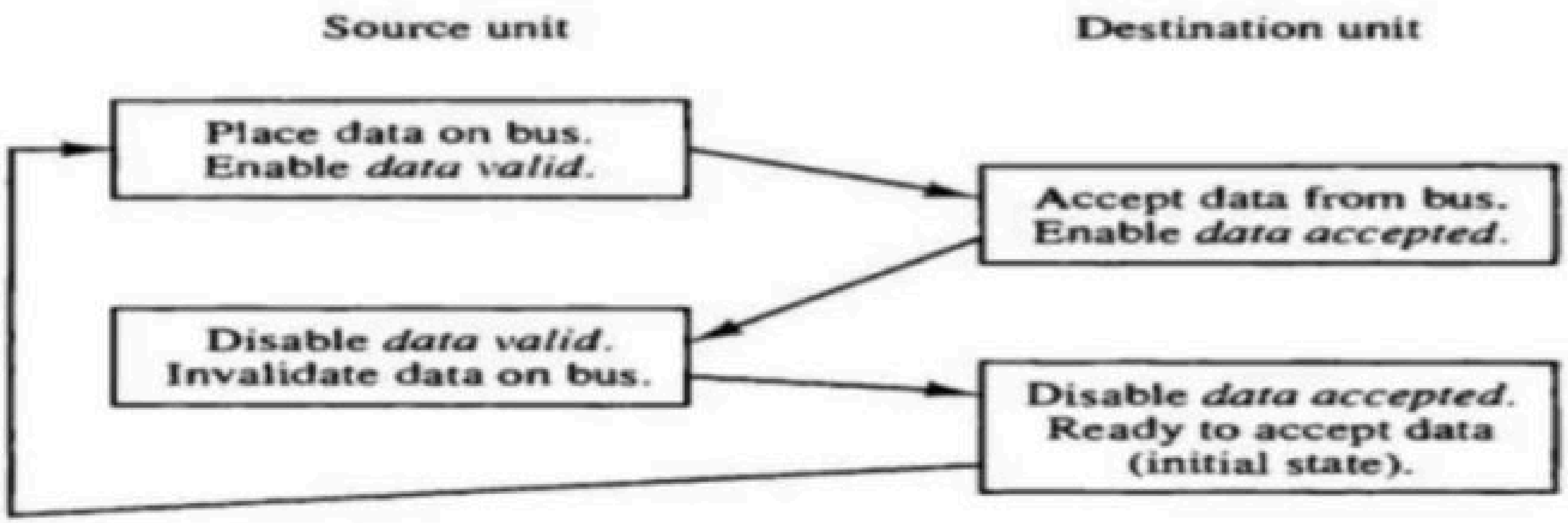
Source Initiated Transfer



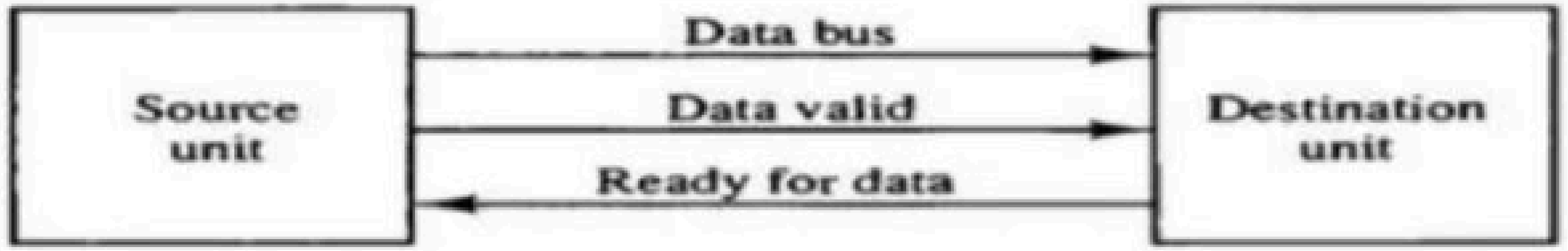
(a) Block diagram



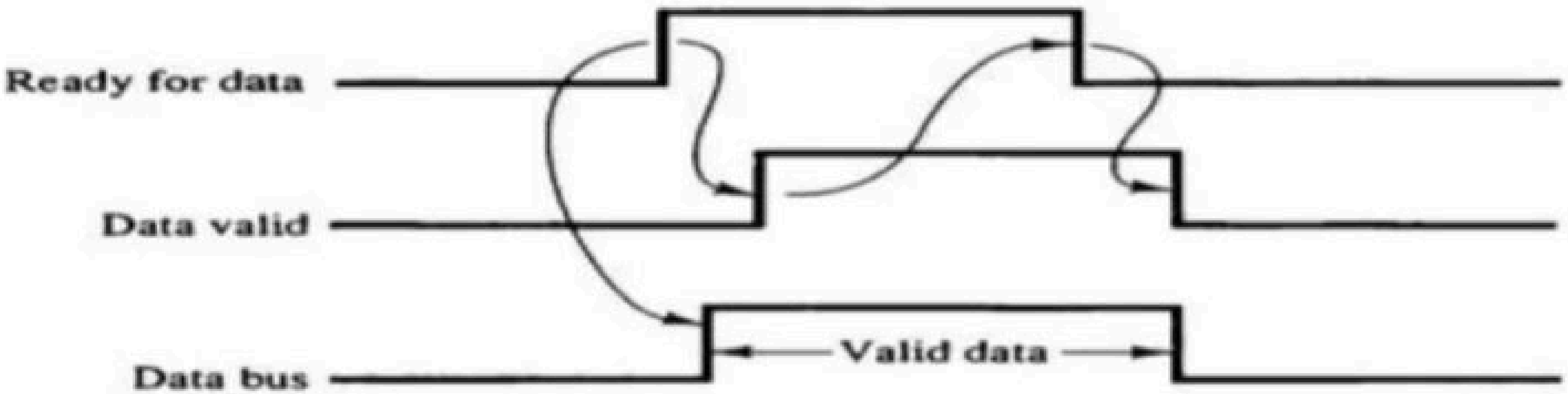
(b) Timing diagram



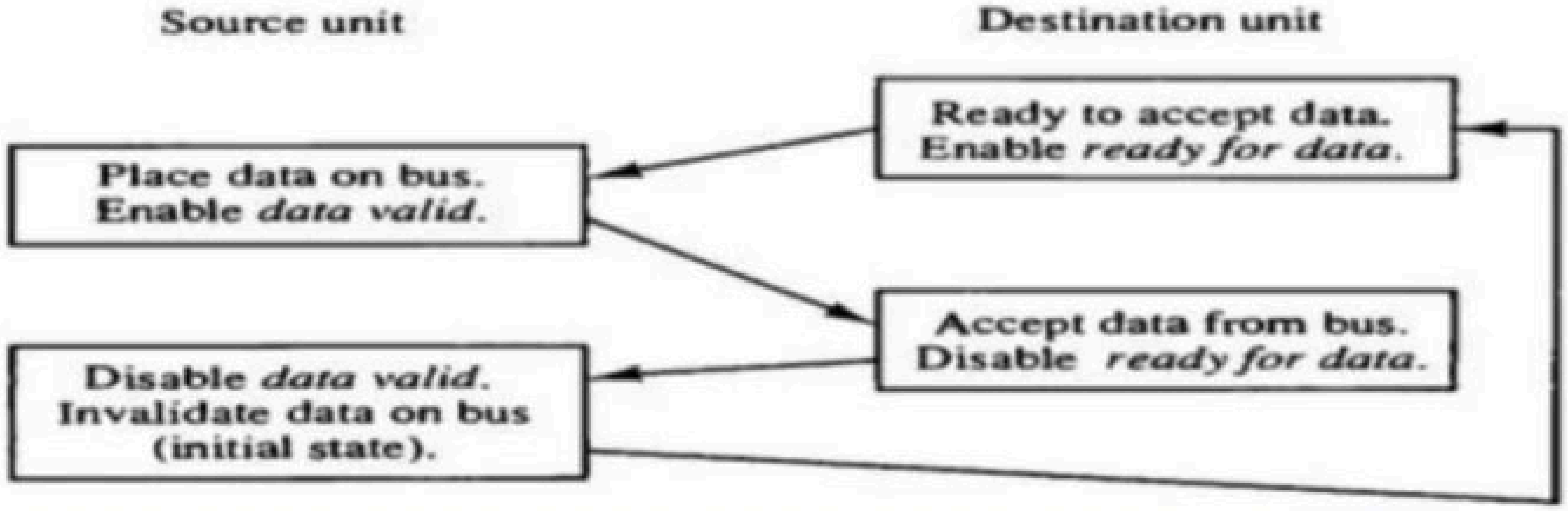
Destination initiated transfer



(a) Block diagram



(b) Timing diagram



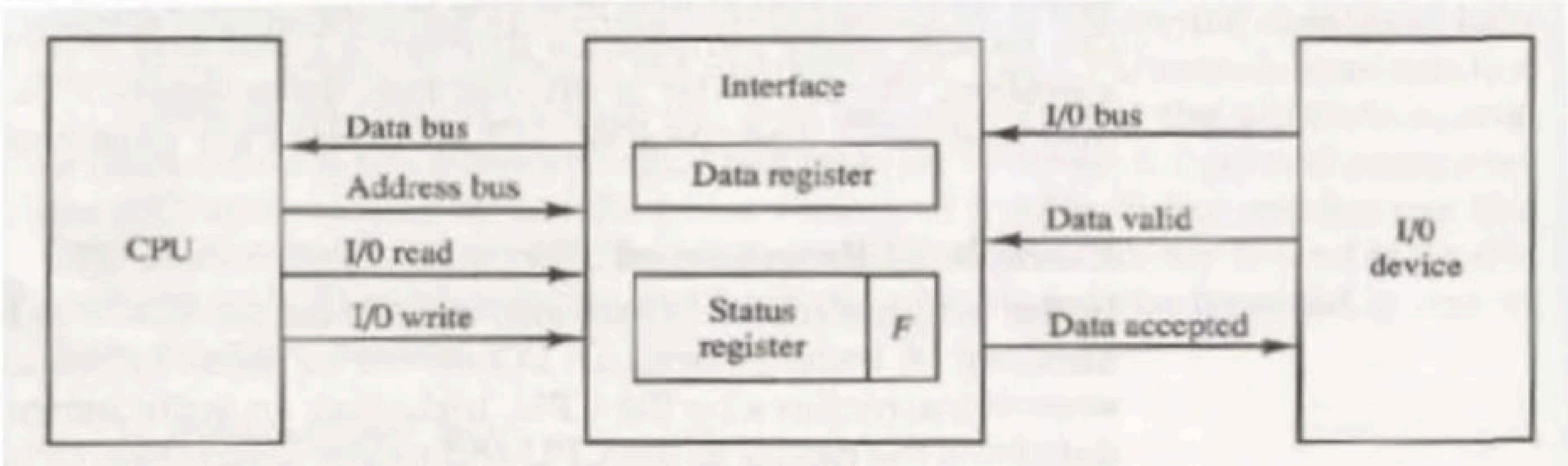
Modes of Data Transfer

Here we deal with the problem that how data communication will take place CPU and i/o device. There are popularly three methods of data transfer: -

- **Programmed i/o**
- **Interrupt initiated i/o**
- **Direct Memory Transfer**

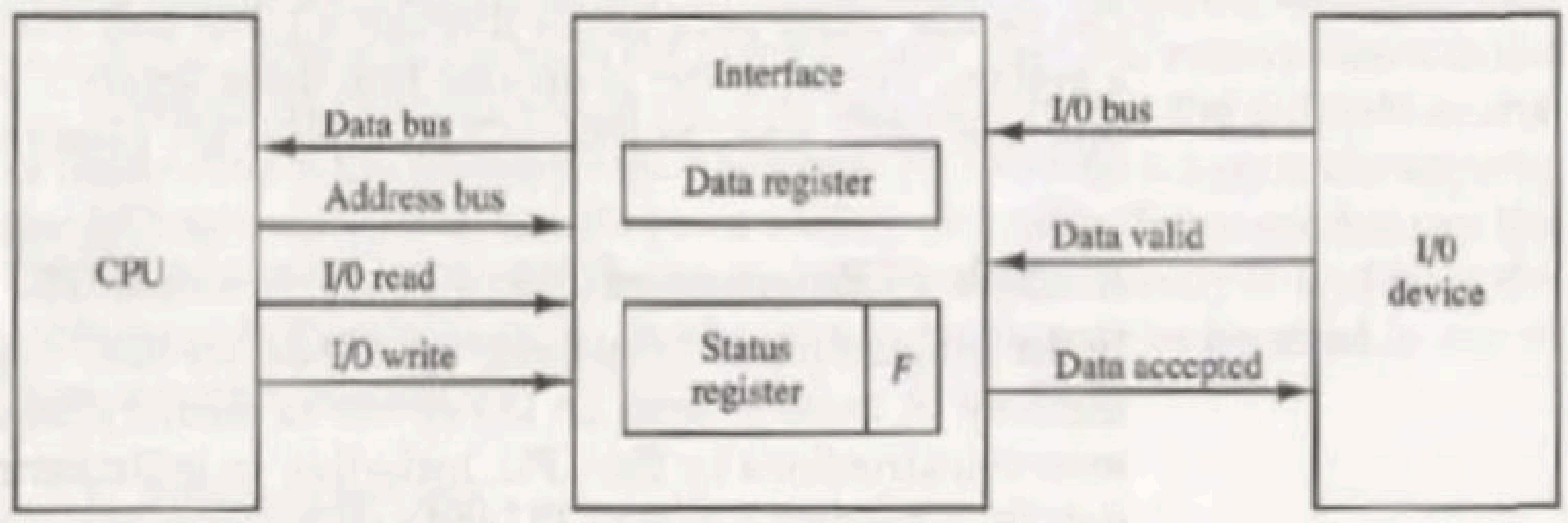
Programmed I/O

- In this i/o device cannot access the memory directly. To complete data transfer number of instructions will be executed out of which input instruction are those which transfer data from device to CPU and store instructions from CPU to memory.



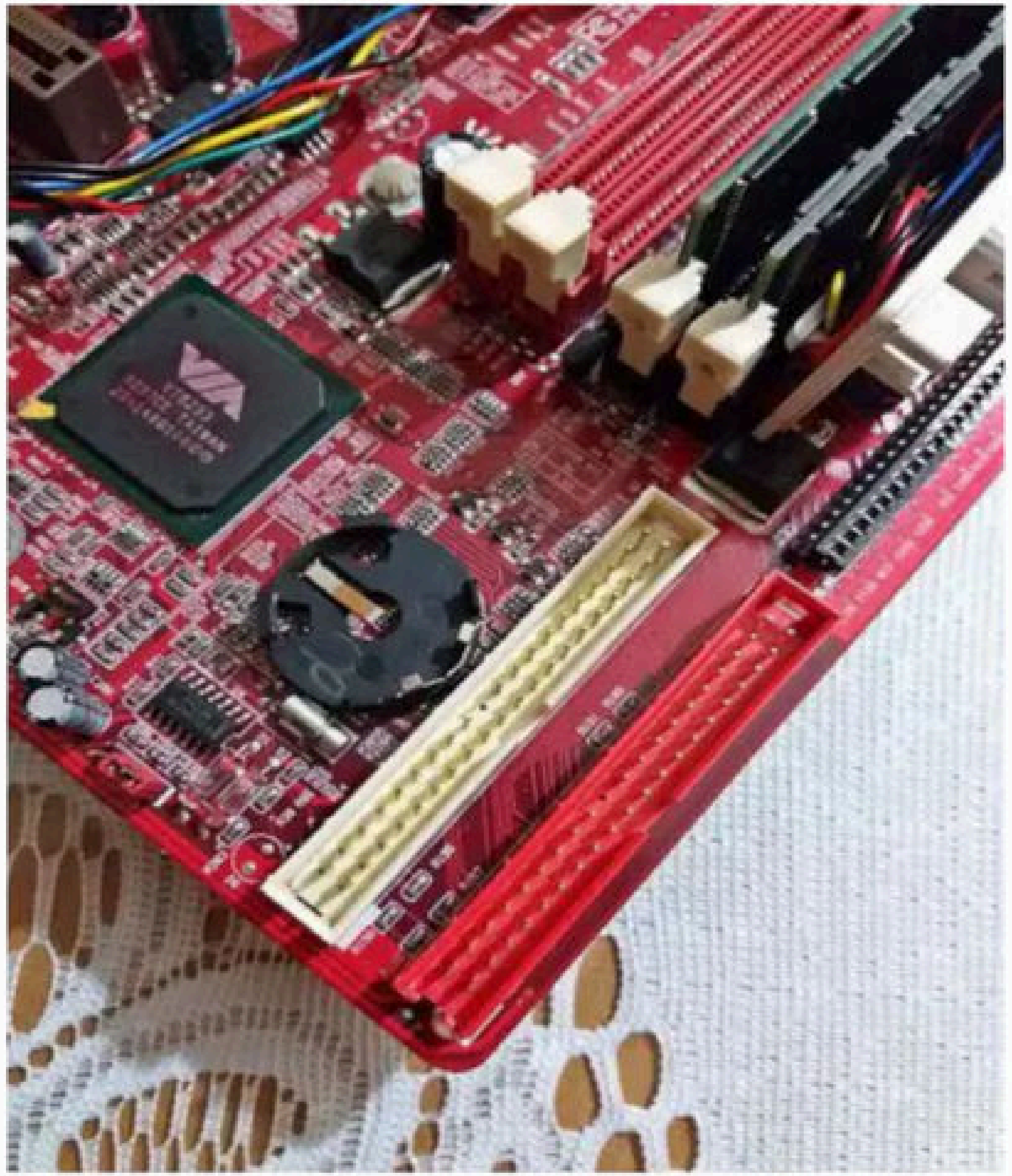
F = Flag bit

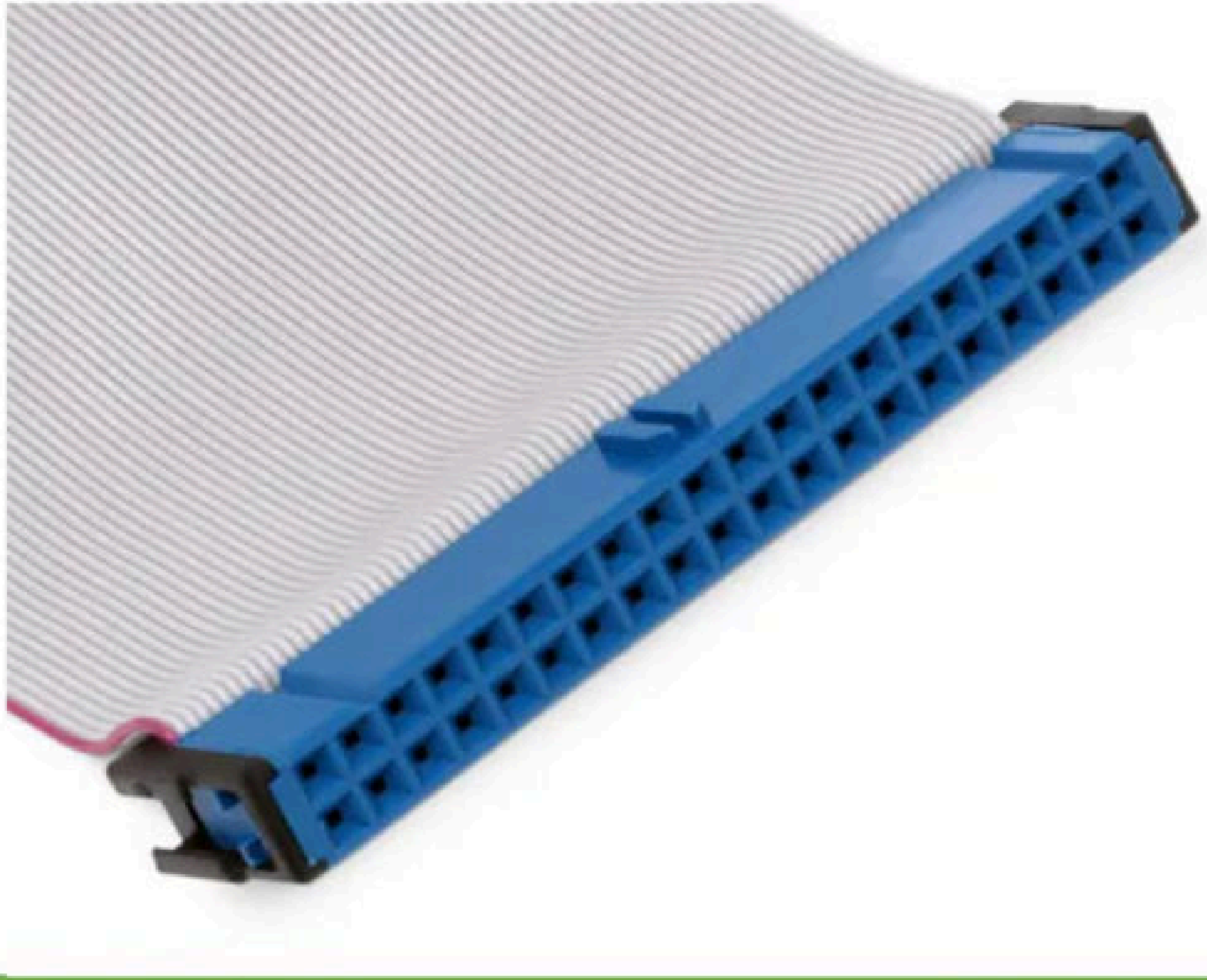
- **Step 1: -** i/o device will put data on the data bus and will enable valid data signal
- **Step 2: -** Interface is continuously sensing for data valid signal and when it receives the signal it will copy data from the data bus into it's data register and set its flag bit to 1 and enable data accepted line.
- **Step 3: -** CPU is continuously monitoring the status register in the programmed mode and as soon as it sees flag bit as 1, it immediately copies data from data register on the data bus and clear flag bit to zero.
- **Step 4: -** Now interface will disable data accepted line to tell i/o device, I am ready for new transitions.
- **Conclusion: -** CPU works in programmed mode or in busy wait mode so no of clock cycles are wasted. It will be difficult to handle multiple i/o device at the same time. It is not appropriate with the high-speed i/o devices.



F = Flag bit

- The best known example of a PC device that uses programmed I/O is the ATA interface
- **Parallel ATA** (**PATA**), originally **AT Attachment**, is an interface standard for the connection of storage devices such as hard disk drives, floppy disk drives, and optical disc drives in computers.



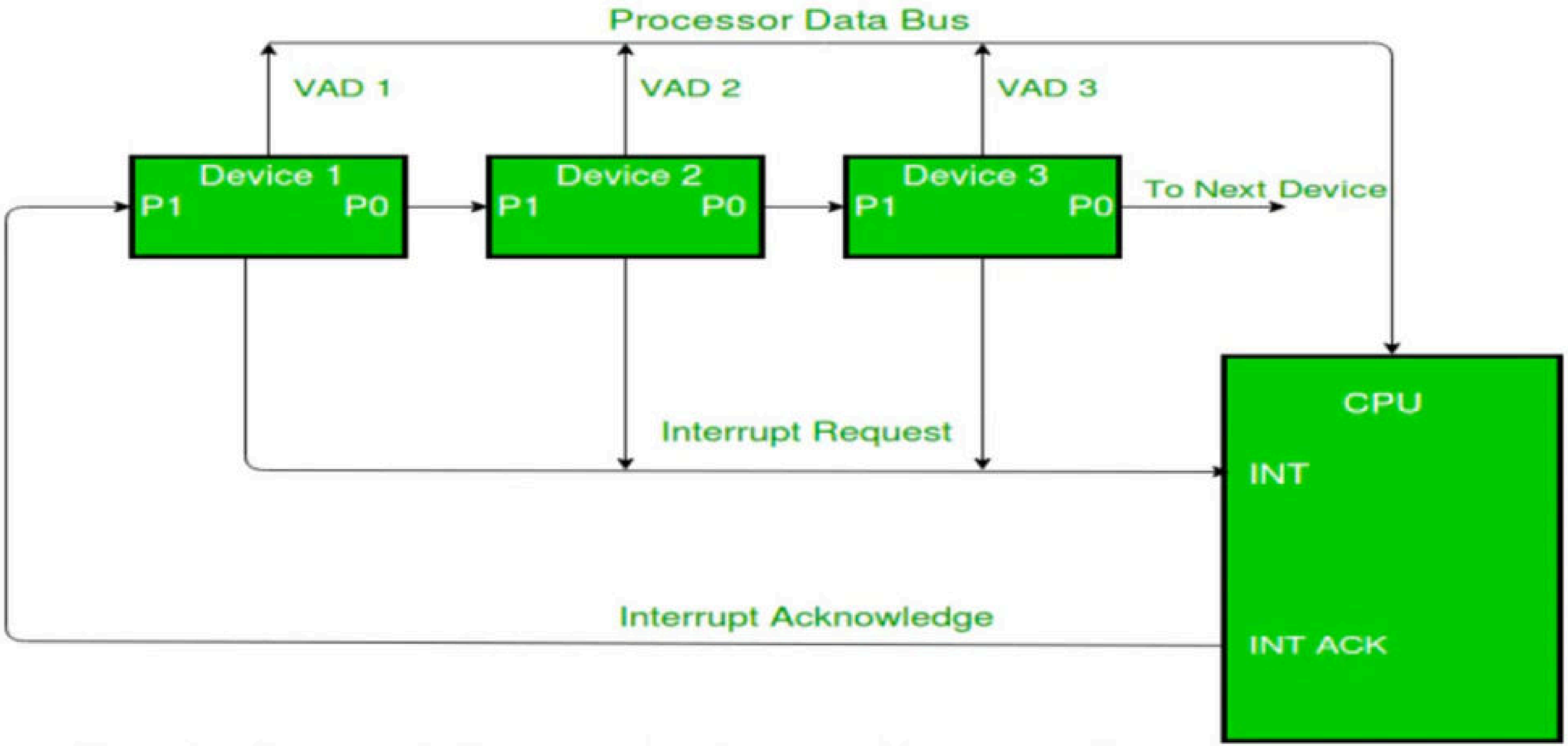


Interrupt initiated i/o

- **In** this method I/o device interrupt CPU when it is ready for data transfer.
- CPU keep executing instructions and after executing one instruction and before starting another instructions CPU wait and see if there is interrupt or not and if there is interrupt then takes a decision whether CPU should entertain this interrupt or continue with the execution.
- **Note: -** instruction is always absolute in nature and there is nothing like partial execution of instruction.
- The method of handling interrupts of different i/o devices are different, therefor every device has a program or routine called ISR (interrupt service routine) which tell CPU how the interrupt should be managed (and it saves CPU time).

- Interrupt can be of two types: -
    - **Non-vectored interrupt: -** Here there is a mutual understanding between CPU and device that where this routine is stored in memory (high priority device)
    - **Vectored interrupt: -** Some interrupts may be vectored where the interrupting device will also tell the address that where this routine is stored in the memory.
- It is possible that different i/o devices interrupt at the same time. Now CPU must have a priority decision that which interrupt should be service first and which should be service later.

- **H/w solution: -** It can be serial or parallel, serial solution is known as Daisy Channing is a h/w solution which is used to decide priority among different i/o devices (VAD- vector address device)



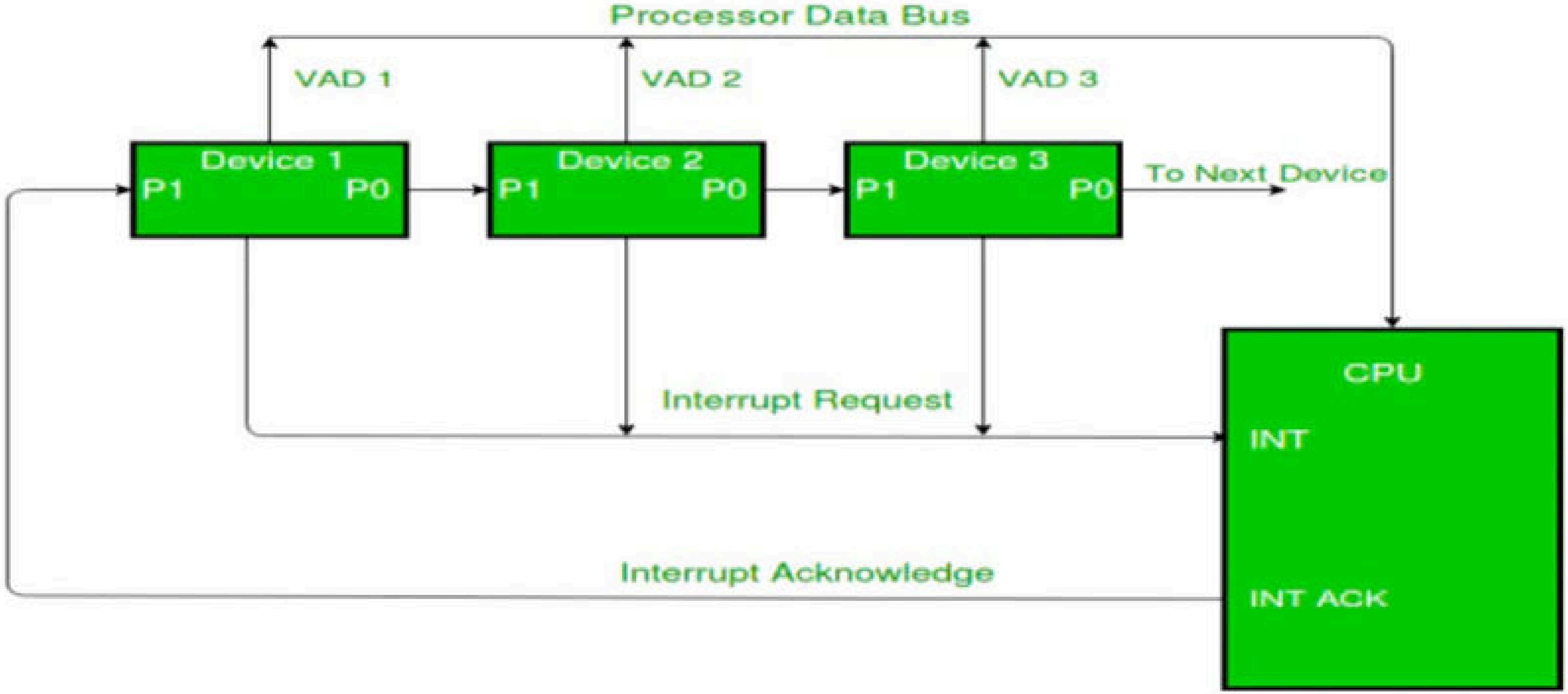
- Out of all possible devices 1 or more device may send an interrupt with a common line.
- When CPU completes 1 instruction and check interrupt in line and found an interrupt, then CPU will enable interrupt acknowledgment line to 1.

- The i/o device placed first in the arrangement will always get acknowledgement first. If it wants to perform i/o then it will put 0 on priority outline and will put the address of its interrupt service routine (ISR) on the vector address line.

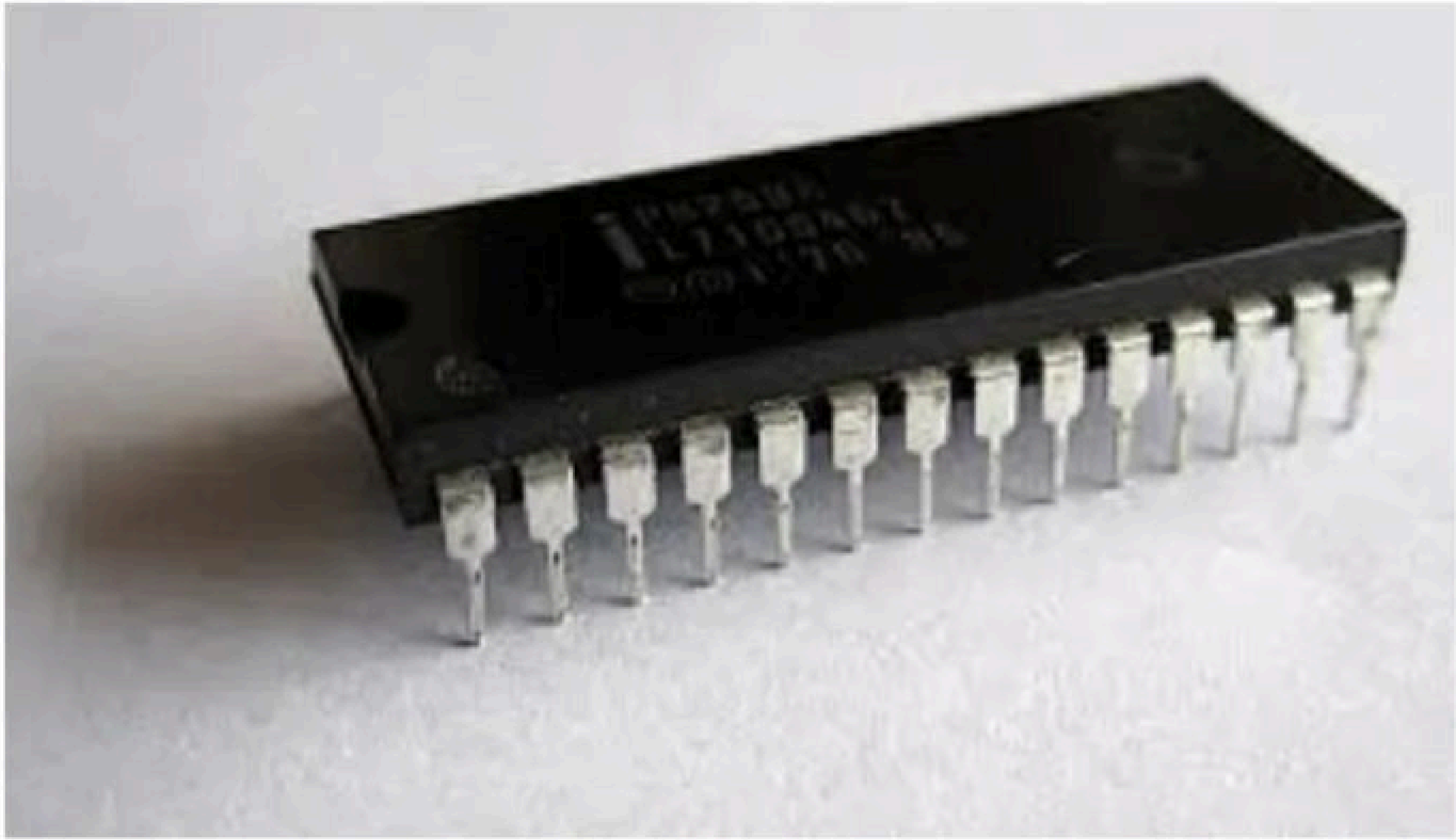
- If device do not want to perform i/o then will set priority output as 1 and will give chance to the second device, and the processor continue.

- **Advantage:** - very simple, easy to use, easy to understand, relatively fast.

- **Disadvantage:** - here priority fixed and even in case of requirement we cannot change it.



- <u>Intel 82C59A</u>



Q The following are some events that occur after a device controller issues an interrupt while process L is under execution. **(GATE-2018) (2 Marks)**

(P) The processor pushes the process status of L onto the control stack.

(Q) The processor finishes the execution of the current instruction.

(R) The processor executes the interrupt service routine.

(S) The processor pops the process status of L from the control stack.

(T) The processor loads the new PC value based on the interrupt.

(A) QPTRS **(B)** PTRSQ **(C)** TRPQS **(D)**QTPRS

Q A CPU handles interrupt by executing interrupt service subroutine __________. **(NET-DEC-2015)**

(a) By checking interrupt register after execution of each instruction

(b) by checking interrupt register at the end of the fetch cycle

(c) whenever an interrupt is registered

(d) by checking interrupt register at regular time interval

Q A CPU generally handles an interrupt by executing an interrupt service routine: **(GATE-2009) (1 Marks)**

a) As soon as an interrupt is raised.

b) By checking the interrupt register at the end of fetch cycle.

c) By checking the interrupt register after finishing the execution of the current instruction.

d) By checking the interrupt register at fixed time intervals.

Q For the daisy chain scheme of connecting I/O devices, which of the following statements is true? **(GATE-1996) (1 Marks)**

a) It gives non-uniform priority to various devices

b) It gives uniform priority to all devices

c) It is only useful for connecting slow devices to a processor device

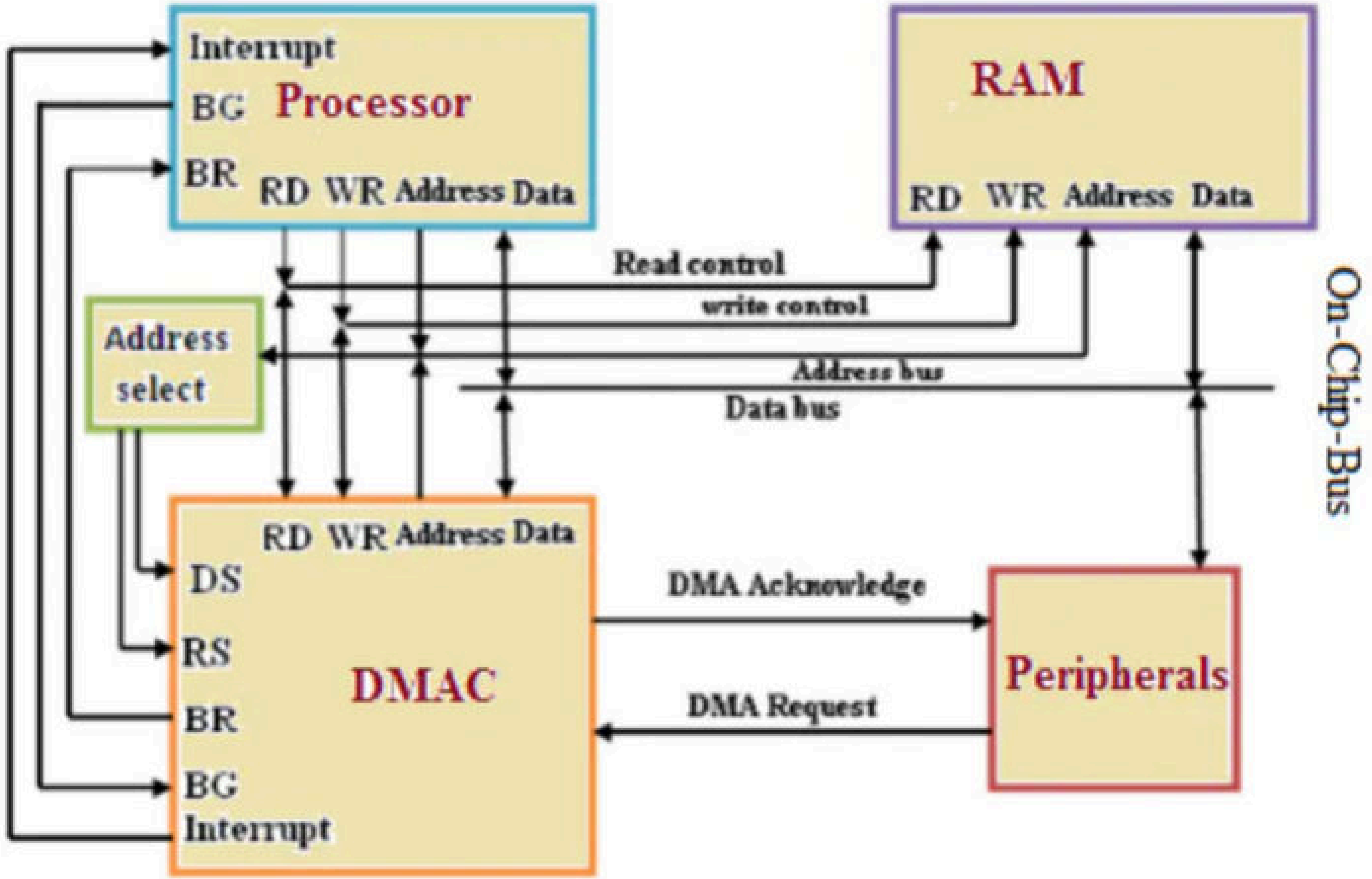
d) It requires a separate interrupt pin on the processor for each device

Q A device with data transfer rate 10 KB/sec is connected to a CPU. Data is transferred byte-wise. Let the interrupt overhead be 4 microsec. The byte transfer time between the device interface register and CPU or memory is negligible. What is the minimum performance gain of operating the device under interrupt mode over operating it under program-controlled mode? **(GATE-2005) (2 Marks)**

(A) 15 **(B)** 25 **(C)** 35 **(D)** 45

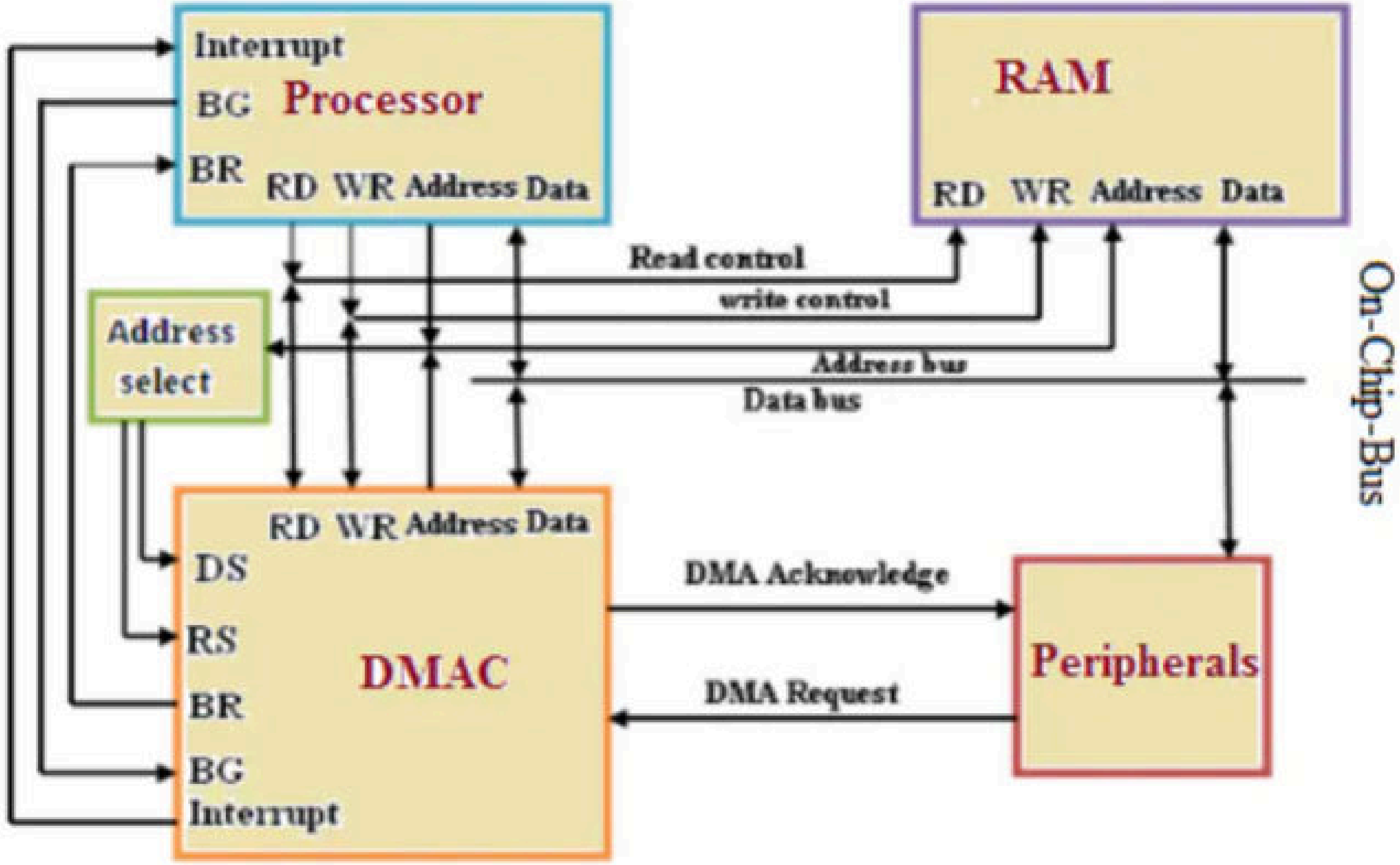
Direct Memory Access (DMA)

- When we want to perform i/o operations then the actual source or destination is either i/o device or memory, but CPU is placed in between just to manage and control the transfer.
- **DMA is** the idea where we use a new device is call DMA controller using which CPU allow DMA controller to take control of system buses and perform direct data transfer either from device to memory or from memory to device.



Sequence of DMA transfer: -

- **Step 1: -** I/O device will send a DMA request to DMA controller to perform an i/o operation.
- **Step 2: -** DMA controller will set interrupt and bus request line to 1.
- **Step 3: -** CPU using address bus will select device and registers and then will initiate i/o address register(location), counter (no of words)
- **Step 4: -** CPU will put on the bus grant line to tell DMA controller, now you are the master of system buses.
- **Step 5: -** now DMA controller will put 1 in DMA acknowledgement and using read/write control lines and address line will perform i/o directly between memory and device.



Mode of transfer: -

- **Burst mode: -** when entire i/o transfer is completed and then control comes back to CPU then it is called burst mode transfer. i.e. with high speed device like magnetic disk.
- **Cycle stealing mode: -** When CPU executes an instruction then normally their could following phases.
    - **IF –** Instruction Fetch
    - **ID –** Instruction Decode
    - **OF –** Operand fetch
    - **IX –** Instruction execute
    - **WB –** write back or store result
- Normally in ID and IE phases CPU don't require system buses and if only in that time control is giving to DMA controller then it is called cycle stealing.

- Many hardware systems use DMA, including disk drive controllers, graphics cards, network cards and sound cards.
- In the original IBM PC (and the follow-up PC/XT), there was only one Intel 8237 DMA controller capable of providing four DMA channels (numbered 0–3).

Q The size of the data count register of a DMA controller is 16 bits. The processor needs to transfer a file of 29,154 kilobytes from disk to main memory. The memory is byte addressable. The minimum number of times the DMA controller needs to get the control of the system bus from the processor to transfer the file from the disk to main memory is _________ **(GATE-2016) (2 Marks)**

Q A hard disk with a transfer rate of 10 Mbytes/ second is constantly transferring data to memory using DMA. The processor runs at 600 MHz, and takes 300 and 900 clock cycles to initiate and complete DMA transfer respectively. If the size of the transfer is 20 Kbytes, what is the percentage of processor time consumed for the transfer operation? **(GATE-2004) (2 Marks)**

(A) 5.0% **(B)** 1.0% **(C)** 0.5% **(D)** 0.1%

Q A DMA controller transfers 32-bit words to memory using cycle stealing. The words are assembled from a device that transmits characters at a rate of 4800 characters per second. The CPU is fetching and executing instructions at an average rate of one million instructions per second. By how much will the CPU be slowed down because of the DMA transfer? **(NET-DEC-2015)**

(a) 0.6% **(b)** 0.12% **(c)** 1.2% **(d)** 2.5%

Q Consider a 32-bit microprocessor, with a 16-bit external data bus, driven by an 8 MHz input clock. Assume that this microprocessor has a bus cycle whose minimum duration equals four input clock cycles. What is the maximum data transfer rate for this microprocessor? **(NET-JUNE-2015)**

a) 8×10^6 bytes/sec

b) 4×10^6 bytes/sec

c) 16×10^6 bytes/sec

d) 4×10^9 bytes/sec

Q On a non-pipelined sequential processor, a program segment, which is a part of the interrupt service routine, is given to transfer 500 bytes from an I/O device to memory.

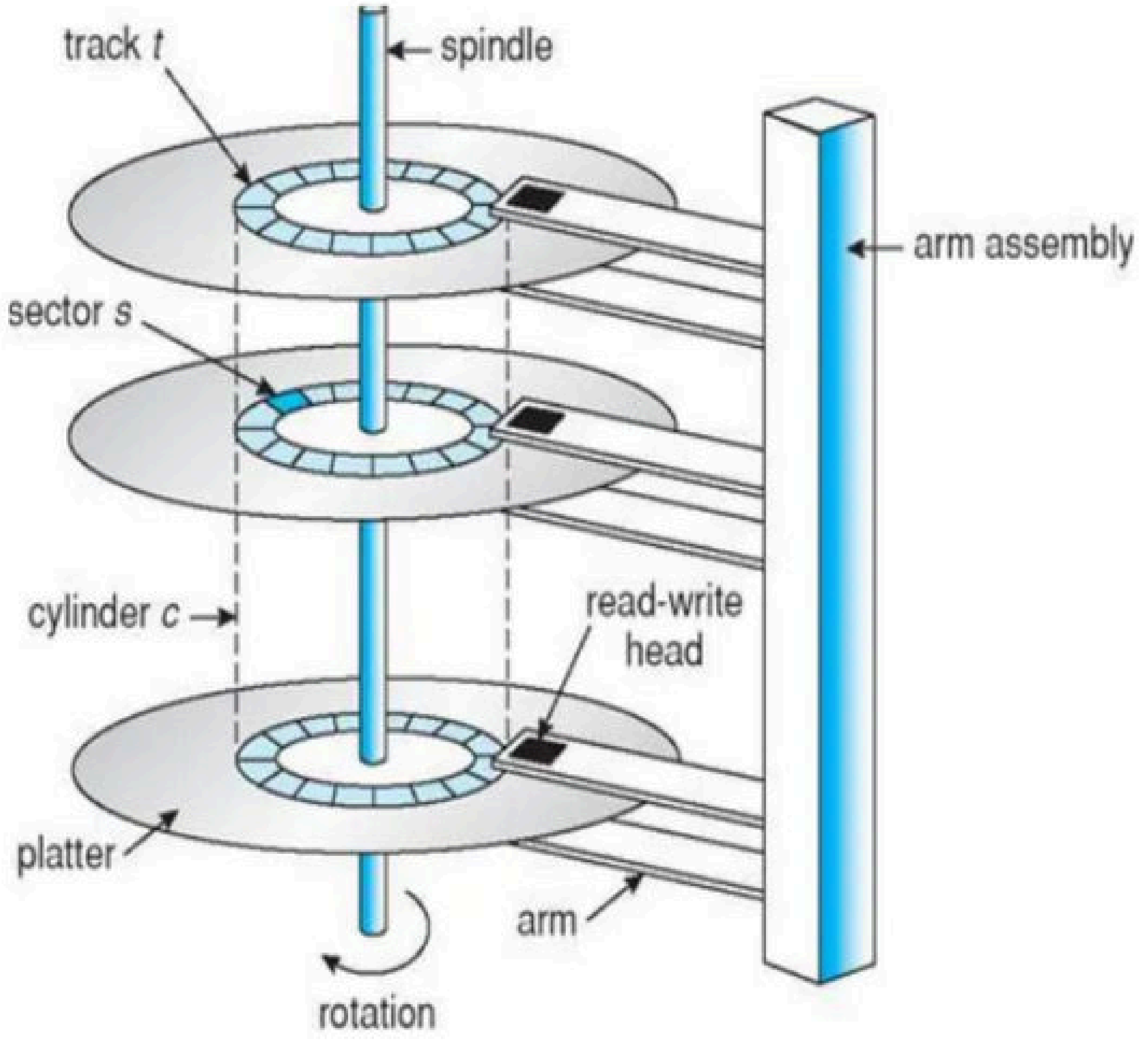
Initialize the address register
Initialize the count to 500

LOOP: Load a byte from device
    Store in memory at address given by address register
    Increment the address register
    Decrement the count
    If count != 0 go to LOOP

Assume that each statement in this program is equivalent to machine instruction which takes one clock cycle to execute if it is a non-load/store instruction. The load-store instructions take two clock cycles to execute. The designer of the system also has an alternate approach of using DMA controller to implement the same transfer. The DMA controller requires 20 clock cycles for initialization and other overheads. Each DMA transfer cycle takes two clock cycles to transfer one byte of data from the device to the memory. What is the approximate speedup when the DMA controller-based design is used in place of the interrupt driven program-based input-output? **(GATE-2011) (2 Marks)**

(A) 3.4 **(B)** 4.4 **(C)** 5.1 **(D)** 6.7

Q.1 Consider a computer system with DMA support. The DMA module is transferring one 8-bit character in one CPU cycle from a device to memory through cycle stealing at regular intervals. Consider a 2 MHz processor. If 0.5% processor cycles are used for DMA, the data transfer rate of the device is _____ bits per second. **(GATE-2021)**


track t
spindle
arm assembly
sector s
cylinder c
read-write
head
platter
arm
rotation

- Total Transfer Time = Seek Time + Rotational Latency + Transfer Time
- **Seek Time:** - It is a time taken by Read/Write header to reach the correct track. (Always given in question)
- **Rotational Latency:** - It is the time taken by read/Write header during the wait for the correct sector. In general, it's a random value, so far average analysis, we consider the time taken by disk to complete half rotation.
- **Transfer Time:** - it is the time taken by read/write header either to read or write on a disk. In general, we assume that in 1 complete rotation, header can read/write the either track, so
- total time will be = (File Size/Track Size) *time taken to complete one revolution.

Q Consider a disk pack with 16 surfaces, 128 tracks per surface and 256 sectors per track. 512 bytes of data are stored in a bit serial manner in a sector. The capacity of the disk pack and the number of bits required to specify a particular sector in the disk are respectively: **(GATE-2007) (1 Marks)**

(A) 256 Mbyte, 19 bits

(B) 256 Mbyte, 28 bits

(C) 512 Mbyte, 20 bits

(D) 64 Gbyte, 28 bit

Q For a magnetic disk with concentric circular tracks, the seek latency is not linearly proportional to the seek distance due to **(GATE-2008) (2 Marks)**

(A) non-uniform distribution of requests

(B) arm starting and stopping inertia

(C) higher capacity of tracks on the periphery of the platter

(D) use of unfair arm scheduling policies

Q If the disk is rotating at 360 rpm, determine the effective data transfer rate which is defined as the number of bytes transferred per second between disk and memory. (track size = 512bytes) **(GATE-1995) (2 Marks)**

Q Consider a typical disk that rotates at 15000 rotations per minute (RPM) and has a transfer rate of 50×10^6 bytes/sec. If the average seek time of the disk is twice the average rotational delay and the controller's transfer time is 10 times the disk transfer time, the average time (in milliseconds) to read or write a 512 byte sector of the disk is ______________ **(GATE-2015) (2 Marks)**

Q Consider a disk pack with a seek time of 4 milliseconds and rotational speed of 10000 rotations per minute (RPM). It has 600 sectors per track and each sector can store 512 bytes of data. Consider a file stored in the disk. The file contains 2000 sectors. Assume that every sector access necessitates a seek, and the average rotational latency for accessing each sector is half of the time for one complete rotation. The total time (in milliseconds) needed to read the entire file is __________. **(GATE-2015) (1 Marks)**

Q A certain moving arm disk storage, with one head, has the following specifications:
Number of tracks/recording surface = 200
Disk rotation speed = 2400 rpm
Track storage capacity = 62,500 bits
The average latency of this device is P ms and the data transfer rate is Q bits/sec. Write the values of P and Q.
(GATE-1993) (2 Marks)

Q A hard disk system has the following parameters: **(GATE-2007) (2 Marks)**

Number of tracks = 500

Number of sectors / track = 100

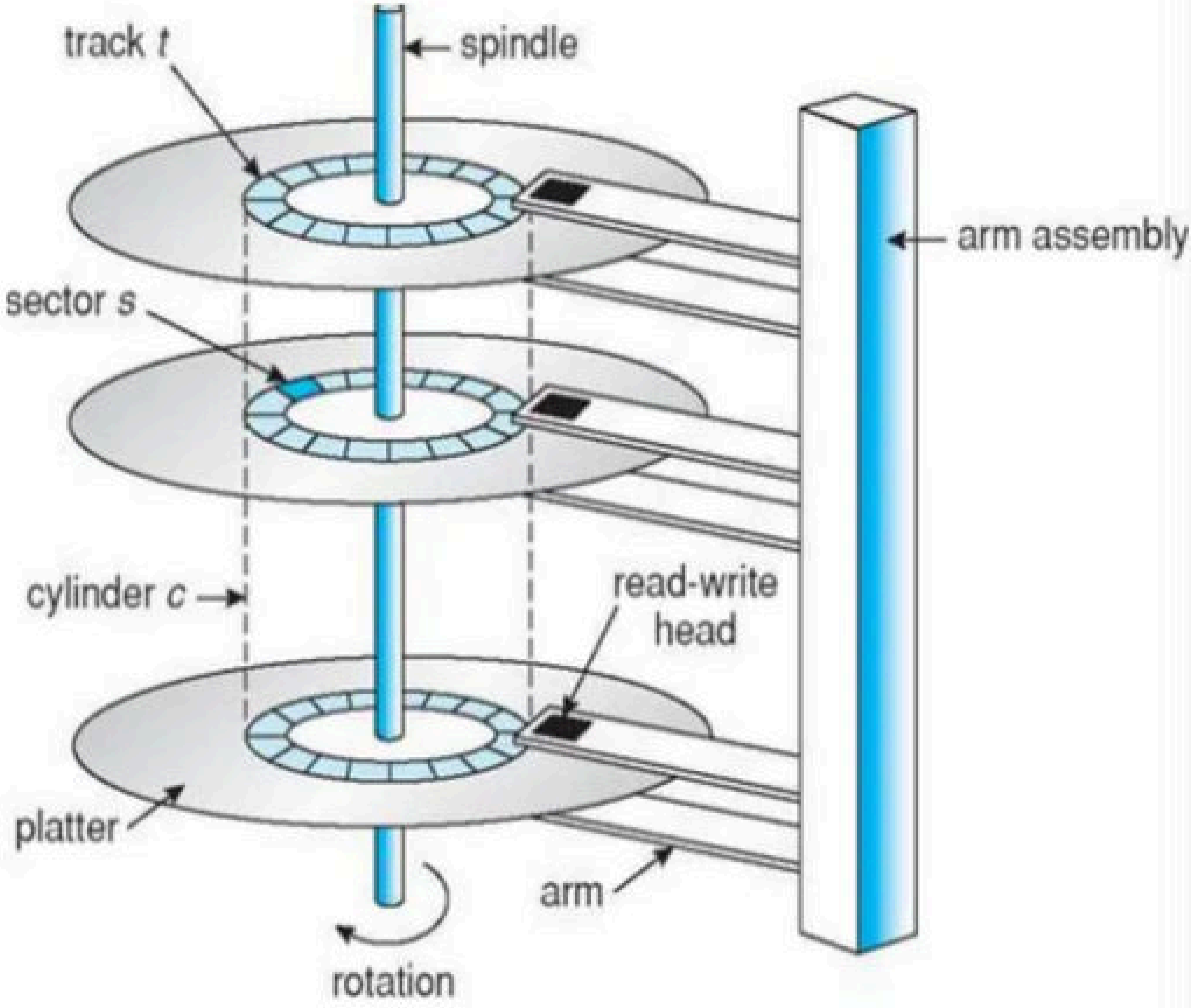
Number of bytes / sector = 500

Time taken by the head to move from one track to adjacent track = 1 ms, Rotation speed = 600 rpm. What is the average time taken for transferring 250 bytes from the disk?

(A) 300.5 ms **(B)** 255.5 ms **(C)** 255.0 ms **(D)** 300.0 ms

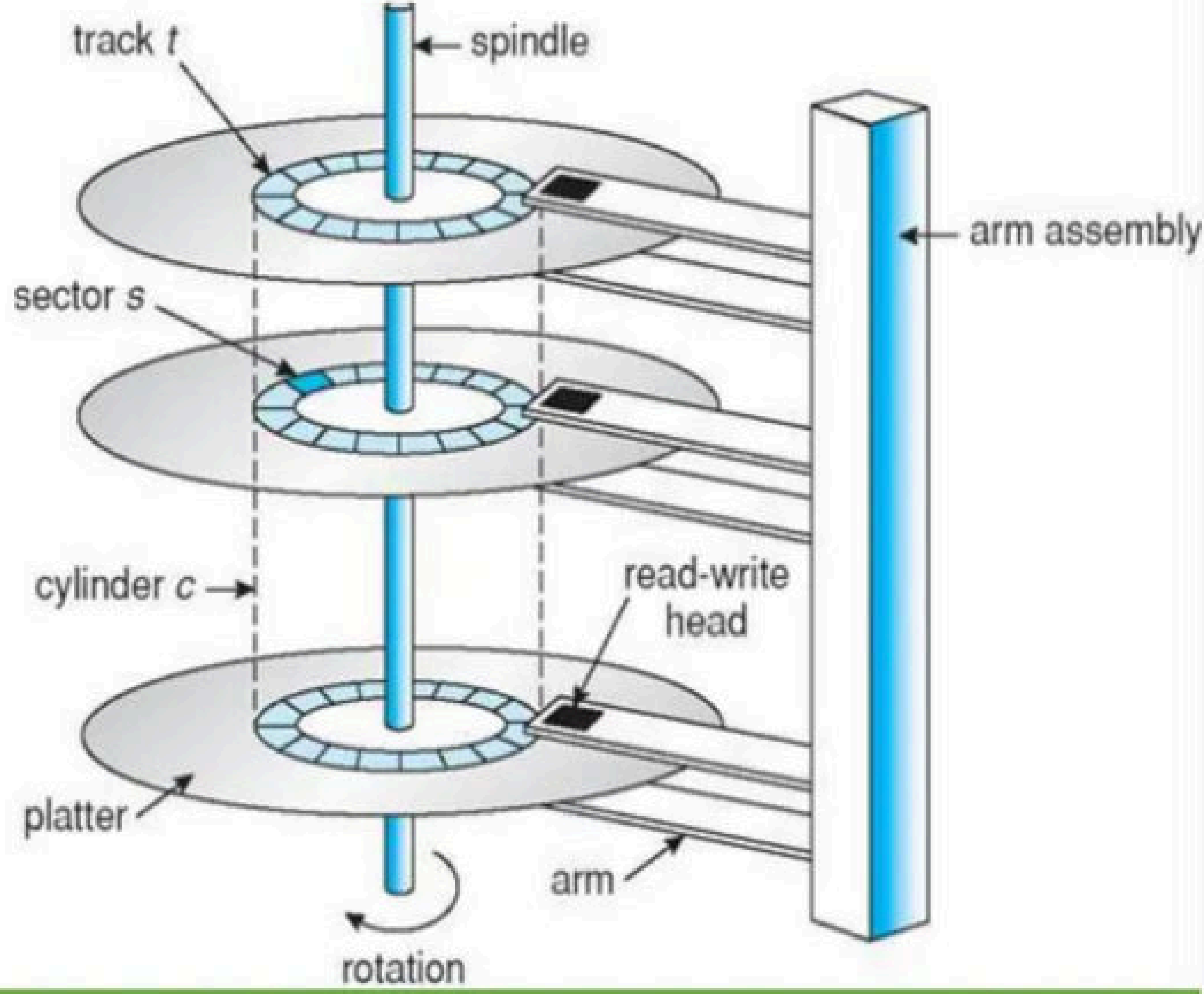
Q A hard disk has 63 sectors per track, 10 platters each with 2 recording surfaces and 1000 cylinders. The address of a sector is given as a triple (c, h, s), where c is the cylinder number, h is the surface number and s is the sector number. Thus, the 0th sector is addressed as (0, 0, 0), the 1st sector as (0, 0, 1), and so on The address <400,16,29> corresponds to sector number: **(GATE-2009) (2 Marks)**

(A) 505035 **(B)** 505036 **(C)** 505037 **(D)** 505038



Q Consider a hard disk with 16 recording surfaces (0-15) having 16384 cylinders (0-16383) and each cylinder contains 64 sectors (0-63). Data storage capacity in each sector is 512 bytes. Data are organized cylinder-wise and the addressing format is <cylinder no., surface no., sector no.>. A file of size 42797 KB is stored in the disk and the starting disk location of the file is <1200, 9, 40>. What is the cylinder number of the last sector of the file, if it is stored in a contiguous manner? **(GATE-2013) (1 Marks)**

(A) 1281 **(B)** 1282 **(C)** 1283 **(D)** 1284



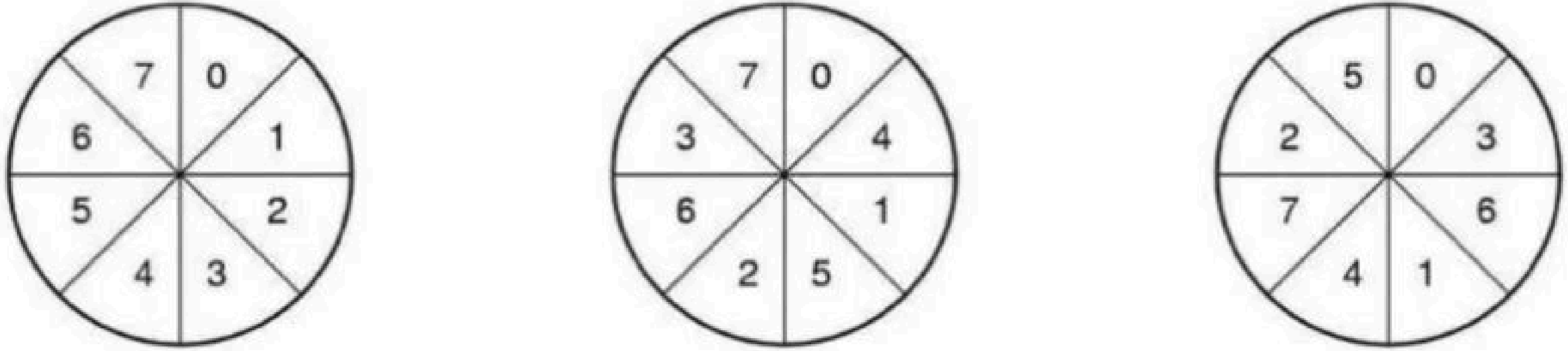
Q Consider a disk with 16384 bytes per track having a rotation time of 16 msec and average seek time of 40 msec. What is the time in msec to read a block of 1024 bytes from this disk? **(NET-DEC-2015)**

a) 57 sec **b)** 49 sec **c)** 48 sec **d)** 17 sec

Q An application loads 100 libraries at start-up. Loading each library requires exactly one disk access. The seek time of the disk to a random location is given as 10 milli second. Rotational speed of disk is 6000 rpm. If all 100 libraries are loaded from random locations on the disk, how long does it take to load all libraries? (The time to transfer data from the disk block once the head has been positioned at the start of the block may be neglected) **(GATE-2011) (2 Marks)**

(A) 0.50 s **(B)** 1.50 s **(C)** 1.25 s **(D)** 1.00 s

- **Single interleaving: -** in 2 rotation we read 1 track
- **Double interleaving: -** in 2.75 rotation we read 1 track



(a) (b) (c)

- No interleaving
- Single interleaving
- Double interleaving